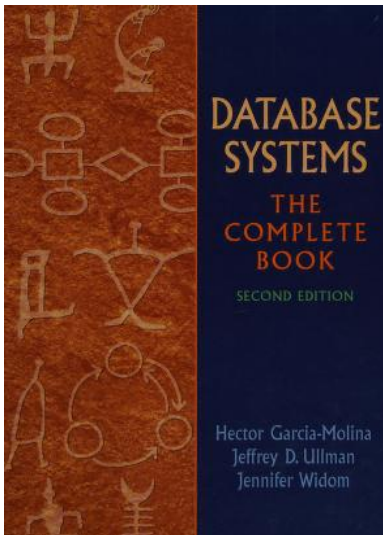


LecW04A Monday: Asynchronous. I'll be recording and put the link in Brightspace. I'll be traveling for a conference.



Conceptual Level | Entity-Relationship²Model

3 Outline: W02B: Thursday



CH04 (2nd Ed.)

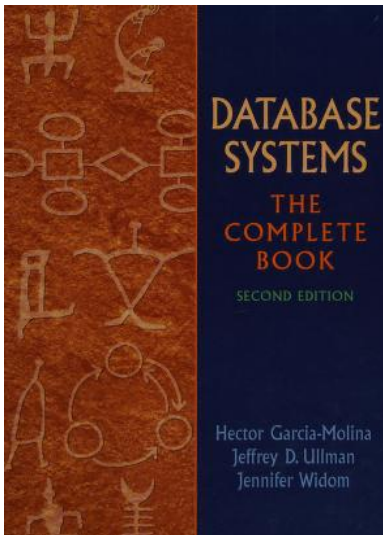
I) Entity

II) Attribute

III) Relationship

III) Extended ER (EER)

Outline: W03A



CH04 (2nd Ed.)

I) Entity

II) Attribute

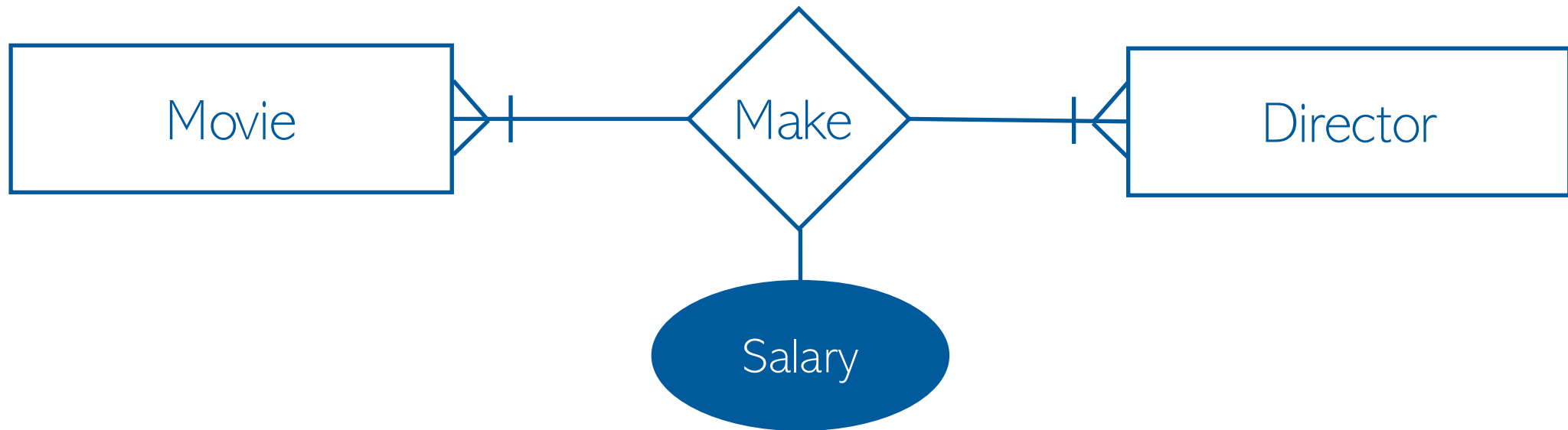
III) Relationship

IIII) Extended ER (EER)

5

Relationship Set (R): **Attribute**

Exist if and only if there is a relationship



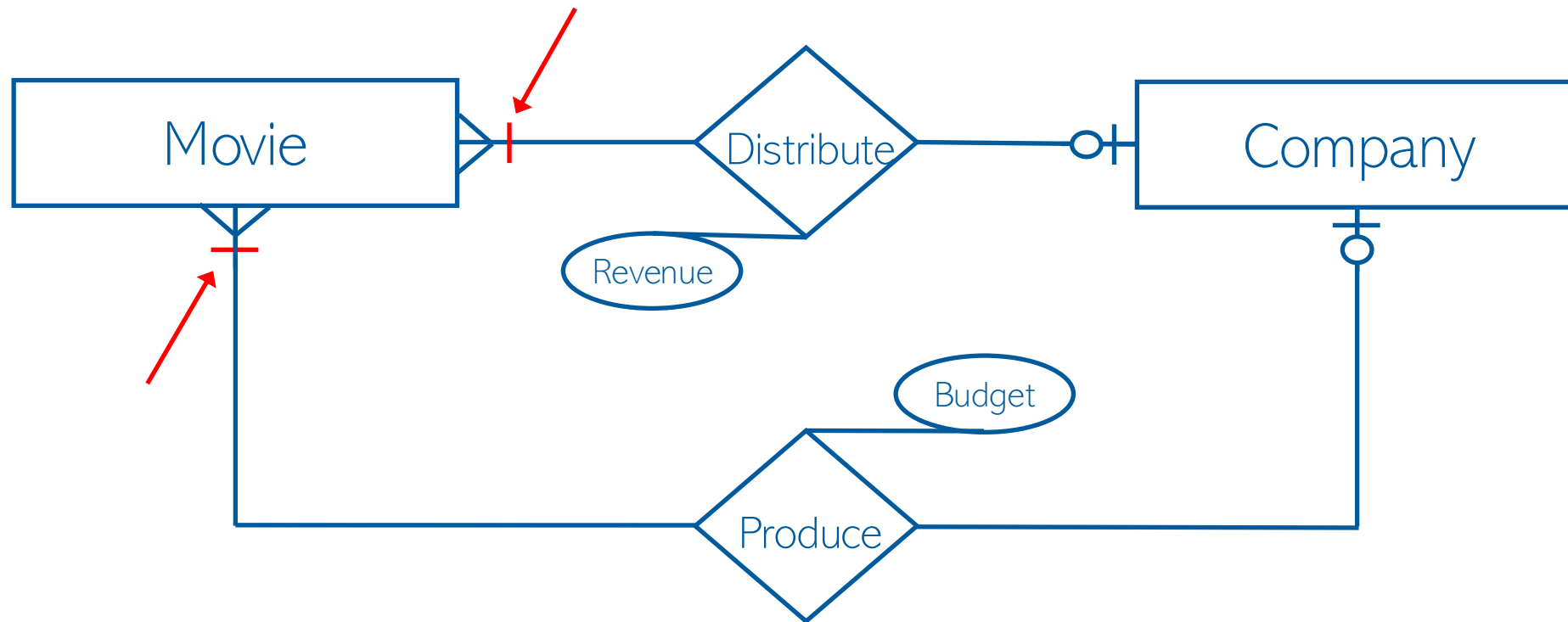
E.g., Salary between a Movie entity and a Director entity.

'Hitchcock' was paid '\$X' to make 'The Birds'.

6

Relationship Set (R): Multiple

A pair of entities of different sets have different types of relationship

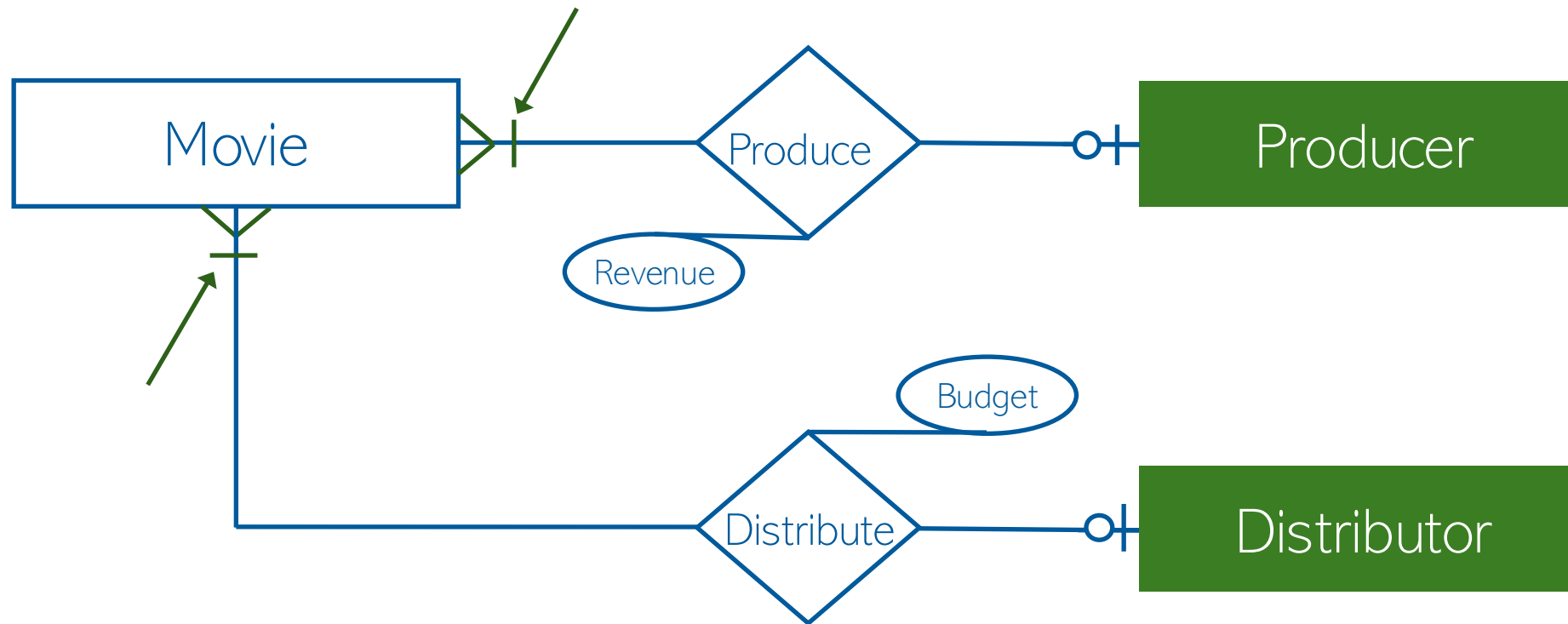


A Company entity **must act as both** Produce and Distribute entity.
'The Birds' was made by 'AIH Inc.' and distributed by 'Universal Pictures'.

7

Relationship Set (R): Multiple

A pair of entities of different sets have different types of relationship



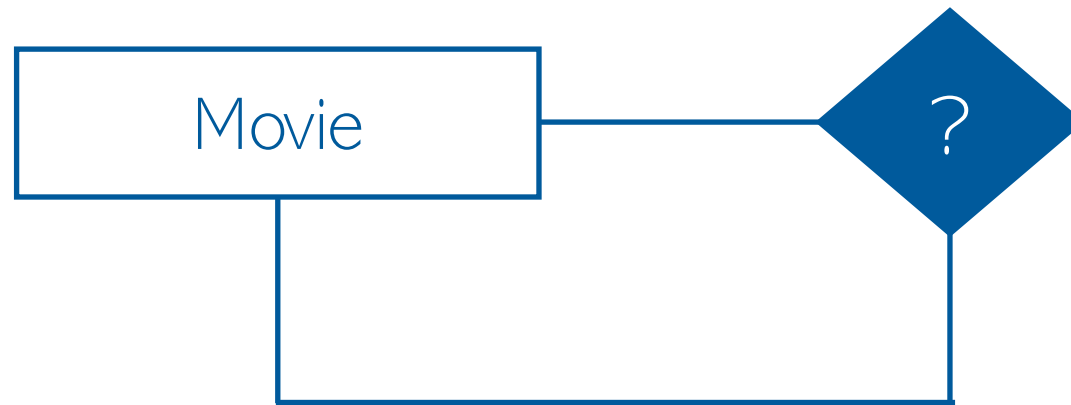
A Producer **must at least** Produce 1 Movie. A Movie can have 0 or 1 Producer.

A Distributor **must at least** Distribute 1 Movie. A Movie can have 0 or 1 Producer.

8

Relationship Set (R): Self (Unary)

A pair of entities of the same entity set have relationship



Can a Movie entity participate in a relationship with another Movie entity?

9

Relationship Set (R): Self (Unary): Multiplicity (Cardinality)

A pair of entities of the same entity set have relationship



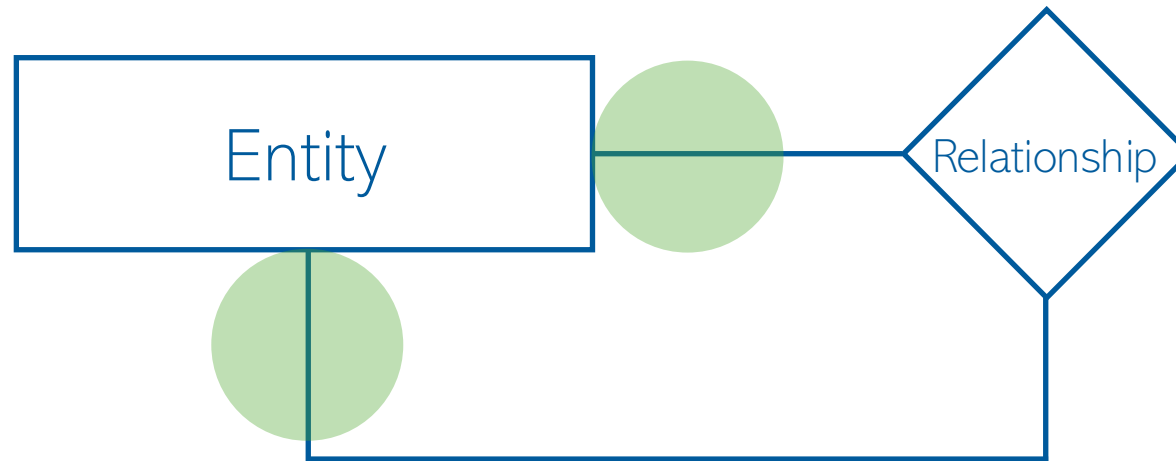
A Movie entity can be SequelOf 1 Movie entity at max.



11

Relationship Set (R): Self (Unary)

A pair of entities of the same entity set have relationship



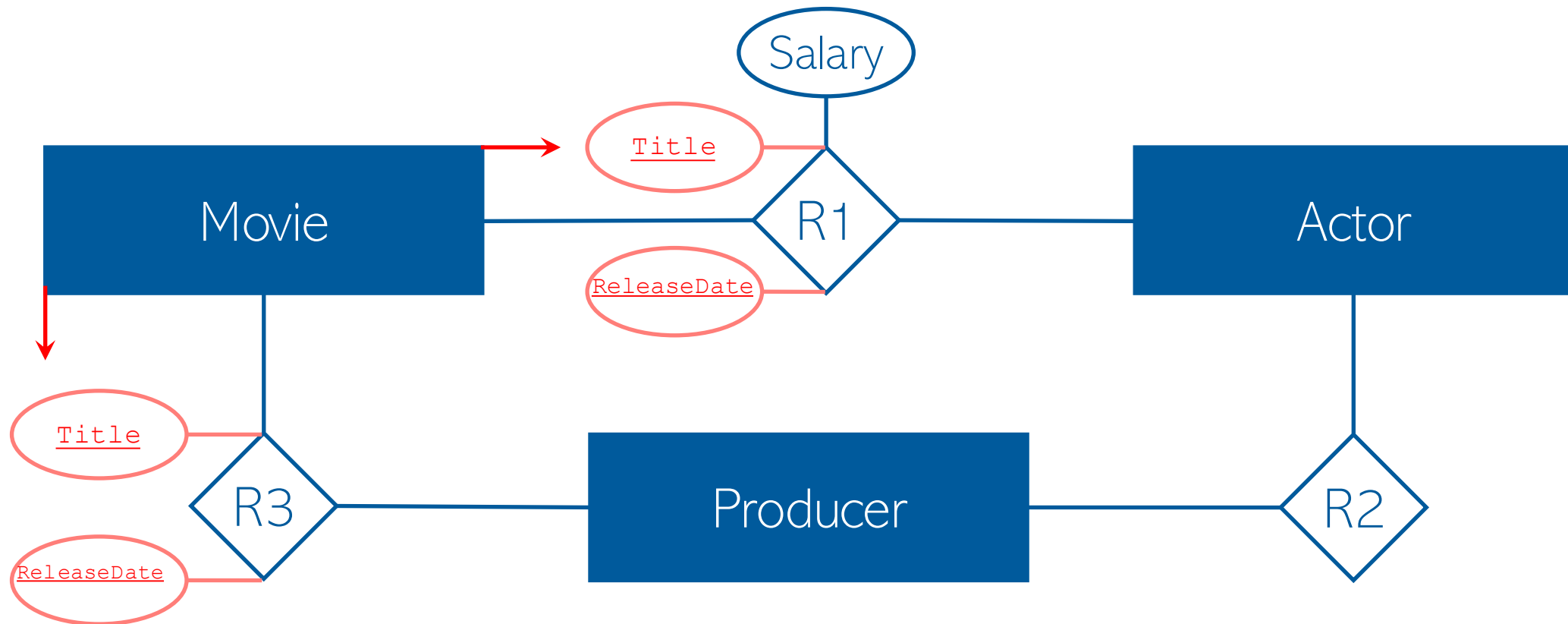
What's the pattern?

Hierarchical relationships among entities of the same entity set.

12

Relationship Set (R): Circular; Poor Design

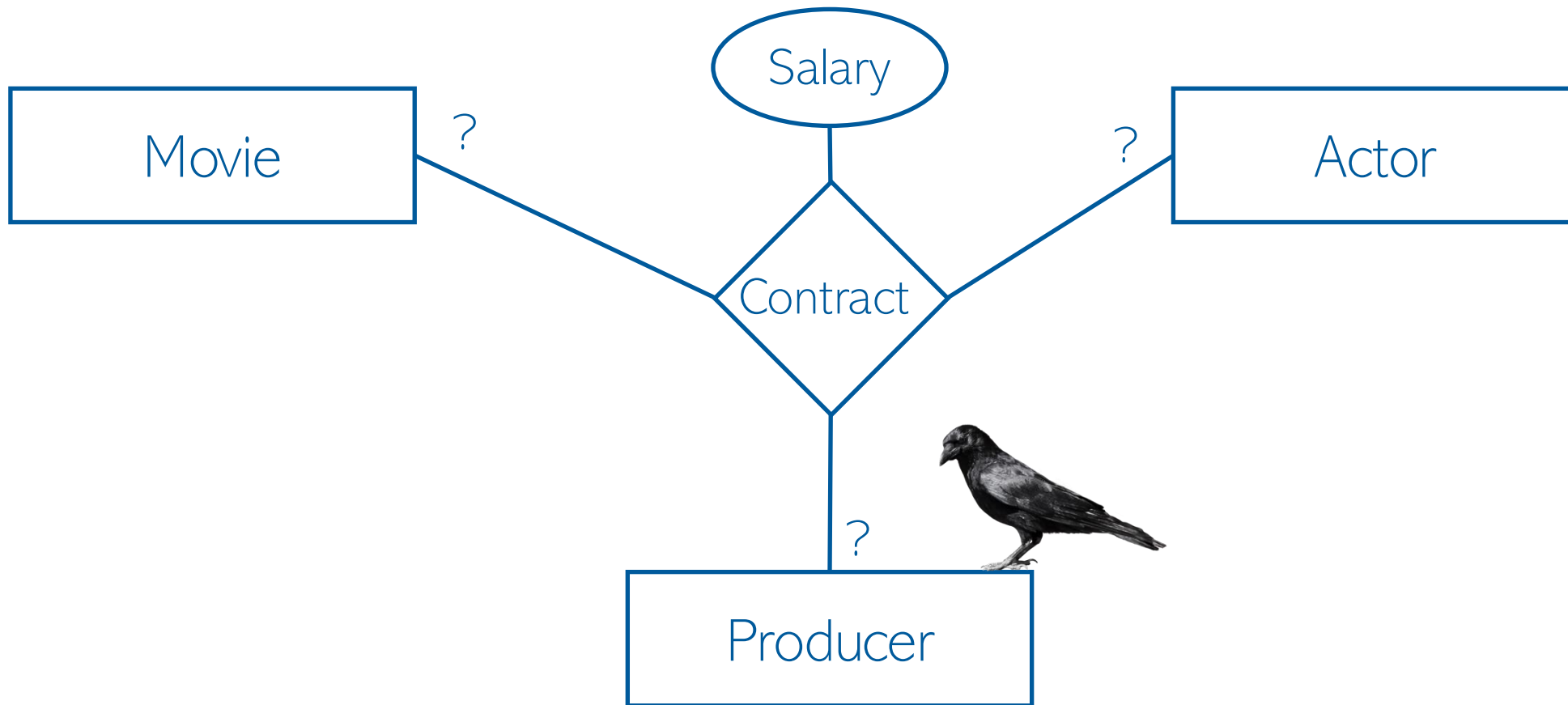
We become biased toward binary relationships ...



13

Relationship Set (R): Ternary

A pair of entities of the same entity set have relationship

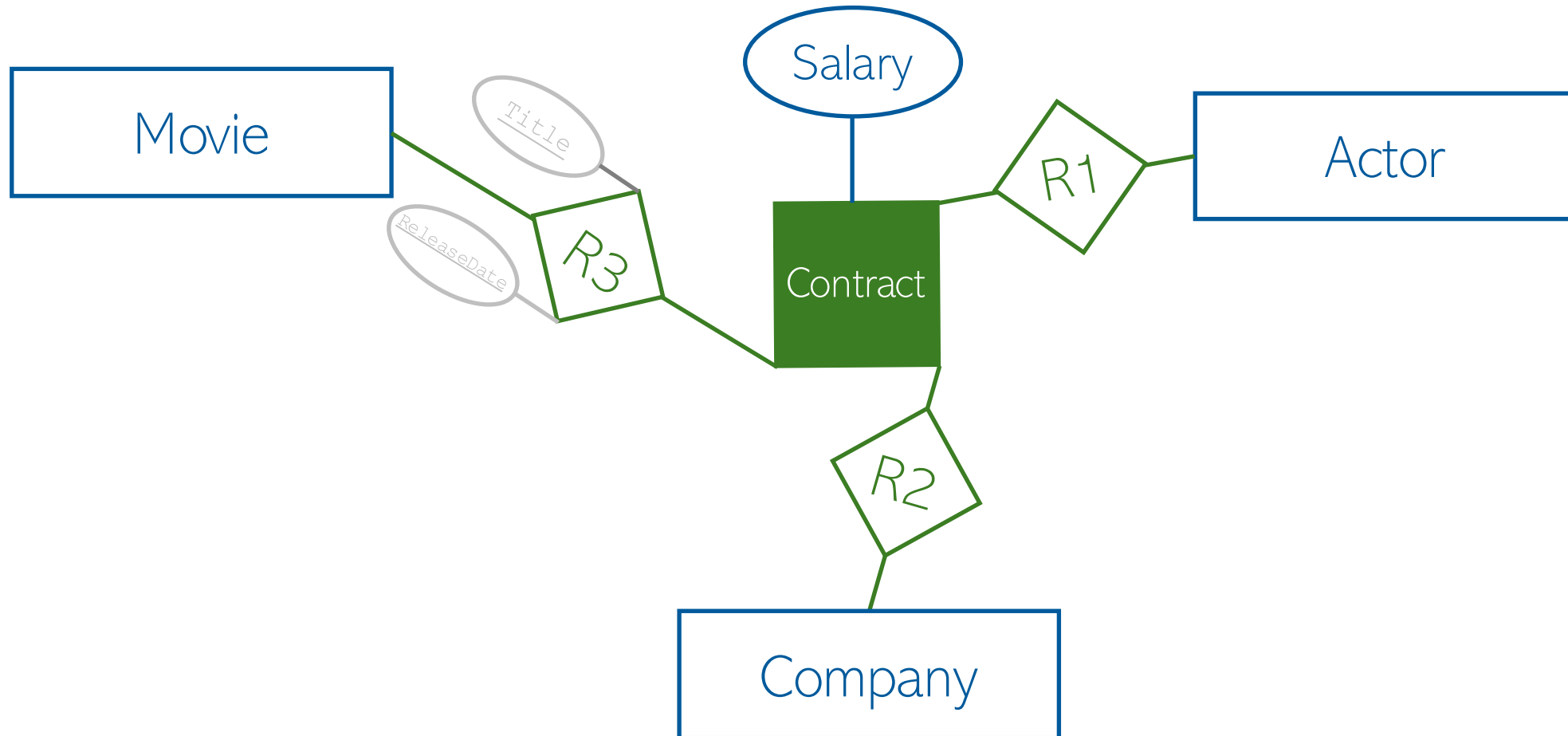


ERDPlus does not support non-binary relationship set/
Either convert to binary, or other tools like www.draw.io

14

Relationship Set (R): Ternary $\rightarrow$ Three Binary

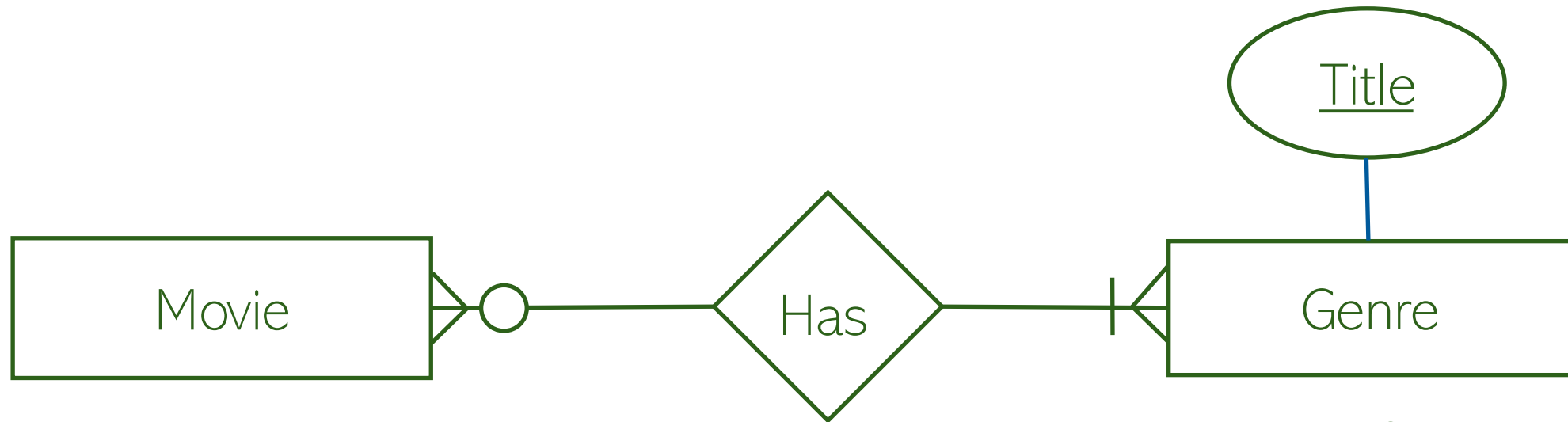
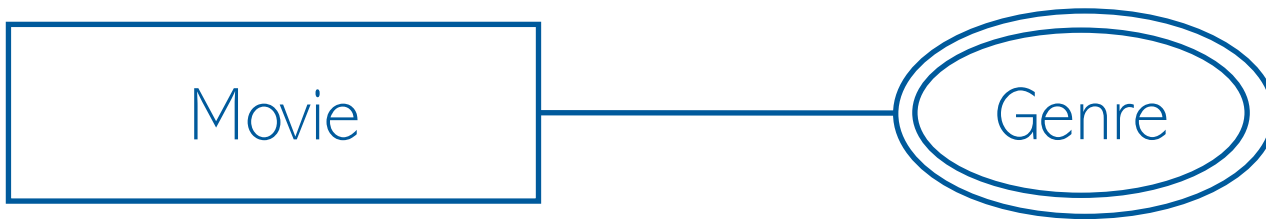
Relationship Set $\rightarrow$ Connecting Entity Set



15

Design Questions

Multivalued Attribute vs. “Has” Relationship

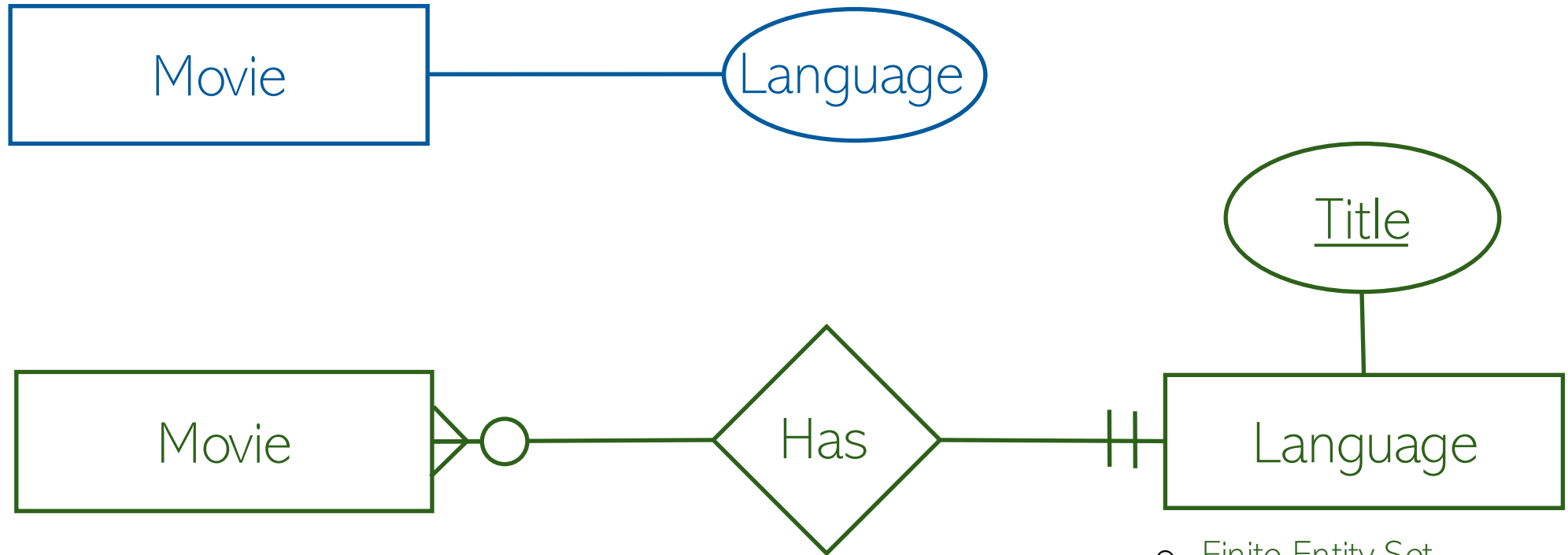


- Finite Entity Set
- Static
- No Multivalued Attribute

16

Design Questions

Multivalued Attribute vs. “Has” Relationship

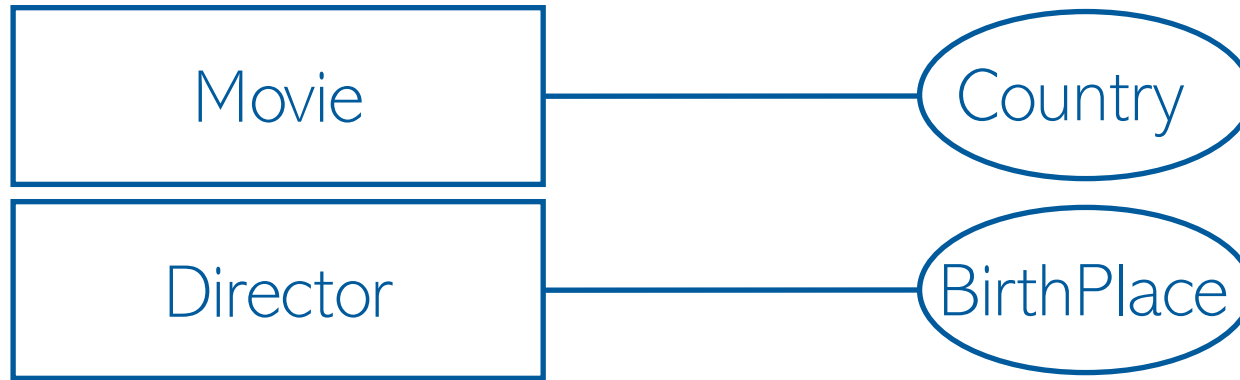


- Finite Entity Set
- Static
- Used in part of the system

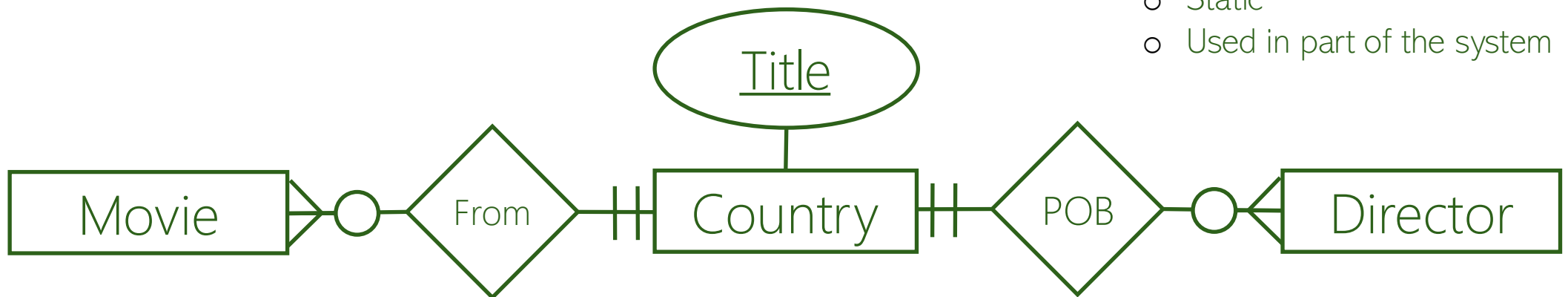
17

Design Questions

Multivalued Attribute vs. “Has” Relationship



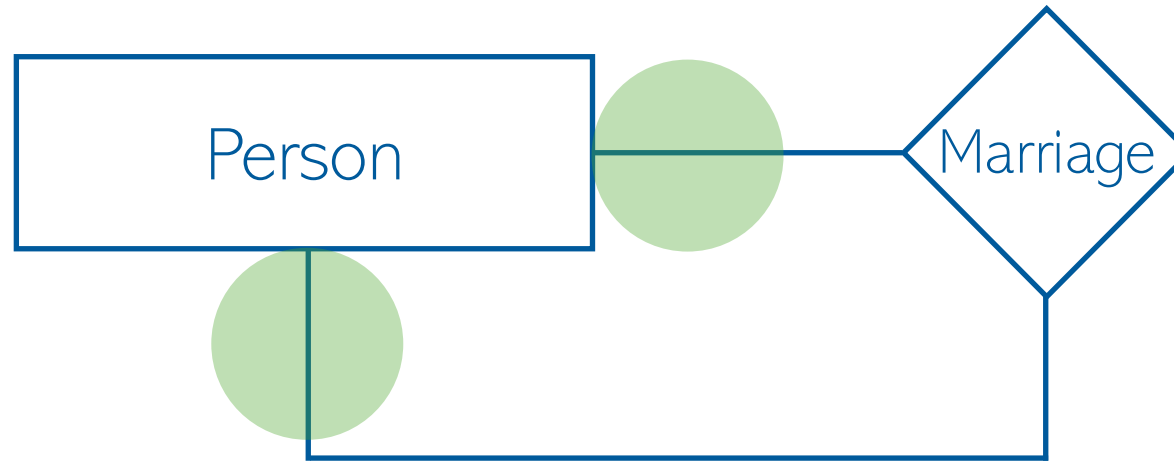
- Finite Entity Set
- Static
- Used in part of the system



18

Relationship Set (R): Practice at Home ...

A pair of entities of the same entity set have relationship



Sologamy *'suh-log-uh-mee'* or *'sah-luh-guh-mee'* (autogamy) is marriage by a person to themselves

Polygyny *'puh·li·juh·nee'* a marriage where one man has more than one wife **at the same time**

Polyandry *'paa·lee·an·dree'* a form of polygamy where a woman has two or more husbands **at the same time**

Polygamy *'puh·li·guh·mee'* is the practice of being married to more than one spouse **at the same time**

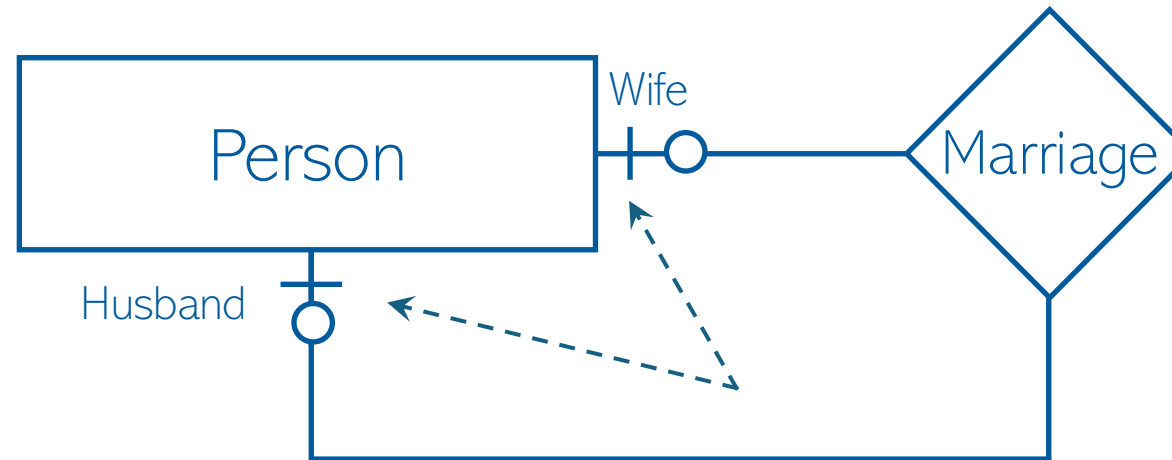
Monogamy *'muh·naa·guh·mee'* is the practice having only one partner at a time

Same Gender/Sex Marriage

19

Relationship Set (R): Marriage

A pair of entities of the same entity set have relationship



Sologamy 'suh-log-uh-mee' or 'sah-luh-guh-mee' (autogamy) is marriage by a person to themselves

Polygyny 'puh·li·juh·nee' a marriage where one man has more than one wife **at the same time**

Polyandry 'paa·lee·an·dree' a form of polygamy where a woman has two or more husbands **at the same time**

Polygamy 'puh·li·guh·mee' is the practice of being married to more than one spouse **at the same time**

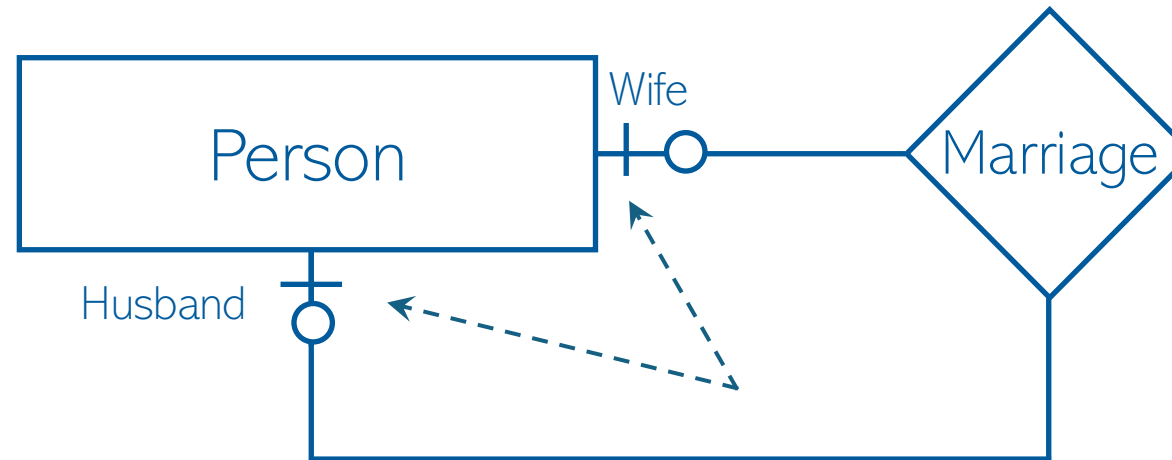
Monogamy 'muh·naa·guh·mee' is the practice having only one partner at a time

Same Gender/Sex Marriage

20

Relationship Set (R): Marriage

A pair of entities of the same entity set have relationship



Sologamy 'suh-log-uh-mee' or 'sah-luh-guh-mee' (autogamy) is marriage by a person to themselves

Polygyny 'puh·li·juh·nee' a marriage where one man has more than one wife **at the same time**

Polyandry 'paa·lee·an·dree' a form of polygamy where a woman has two or more husbands **at the same time**

Polygamy 'puh·li·guh·mee' is the practice of being married to more than one spouse **at the same time**

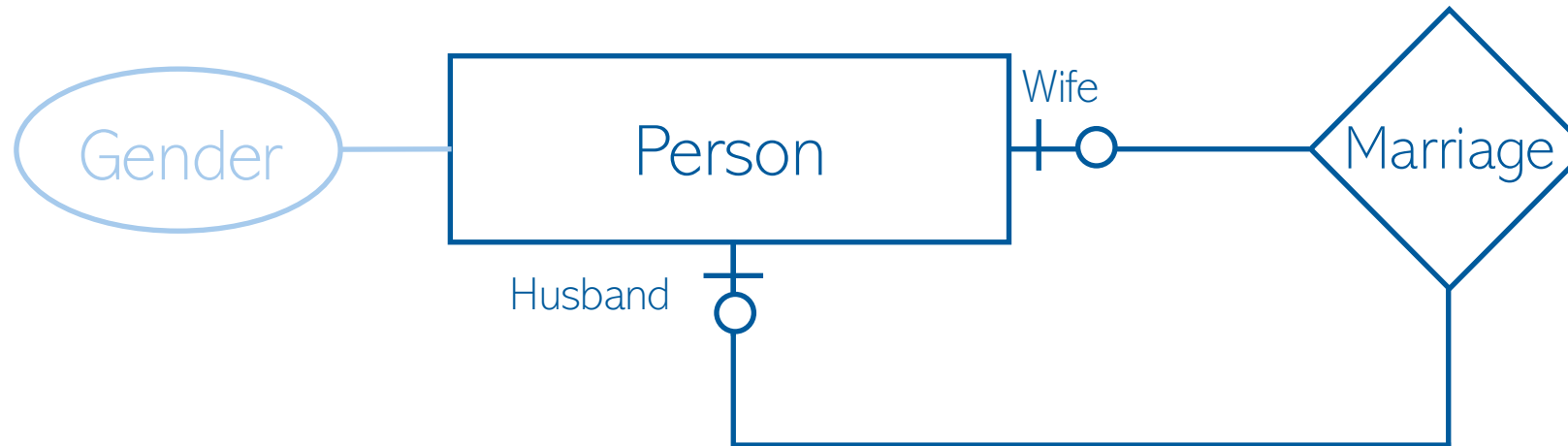
Monogamy 'muh·naa·guh·mee' is the practice having only one partner **at a time**

Same Gender/Sex Marriage

21

Relationship Set (R): Marriage

A pair of entities of the same entity set have relationship



Sologamy 'suh-log-uh-mee' or 'sah-luh-guh-mee' (autogamy) is marriage by a person to themselves

Polygyny 'puh·li·juh·nee' a marriage where one man has more than one wife **at the same time**

Polyandry 'paa·lee·an·dree' a form of polygamy where a woman has two or more husbands **at the same time**

Polygamy 'puh·li·guh·mee' is the practice of being married to more than one spouse **at the same time**

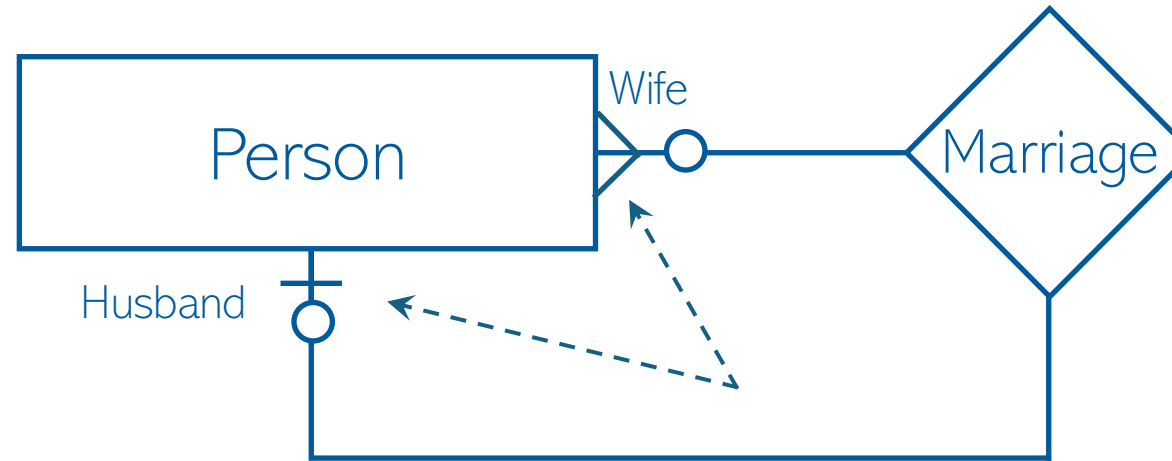
Monogamy 'muh·naa·guh·mee' is the practice having only one partner at a time

Same Gender/Sex Marriage (No need for further change, or add Sex/Gender attribute to Person)

22

Relationship Set (R): Marriage

A pair of entities of the same entity set have relationship



Sologamy *'suh-log-uh-mee'* or *'sah-luh-guh-mee'* (autogamy) is marriage by a person to themselves

Polygyny *'puh·li·juh·nee'* a marriage where one man has more than one wife **at the same time**

Polyandry *'paa·lee·an·dree'* a form of polygamy where a woman has two or more husbands **at the same time**

Polygamy *'puh·li·guh·mee'* is the practice of being married to more than one spouse **at the same time**

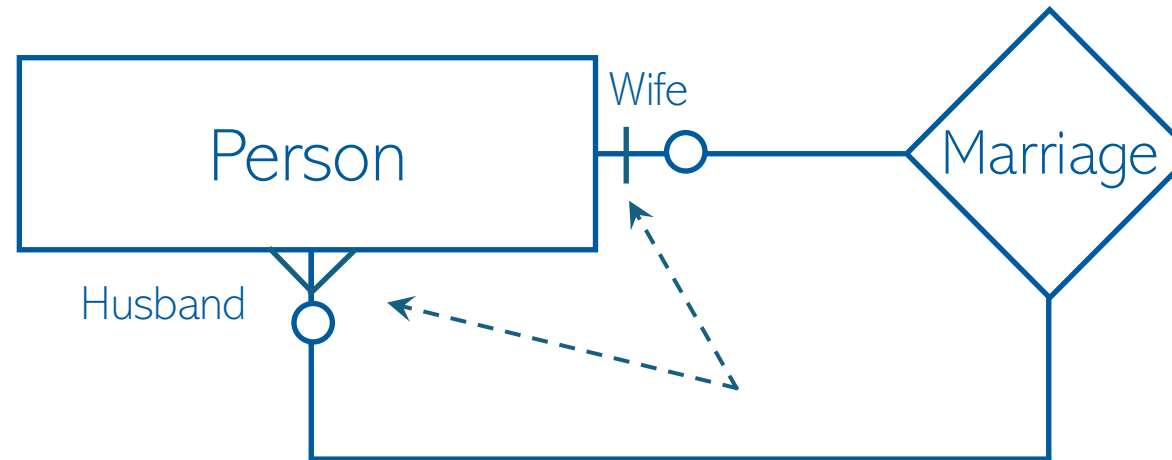
Monogamy *'muh·naa·guh·mee'* is the practice having only one partner at a time

Same Gender/Sex Marriage

23

Relationship Set (R): Marriage

A pair of entities of the same entity set have relationship



Sologamy 'suh-log-uh-mee' or 'sah-luh-guh-mee' (autogamy) is marriage by a person to themselves

Polygyny 'puh·li·juh·nee' a marriage where one man has more than one wife **at the same time**

Polyandry 'paa·lee·an·dree' a form of polygamy where a woman has two or more husbands **at the same time**

Polygamy 'puh·li·guh·mee' is the practice of being married to more than one spouse **at the same time**

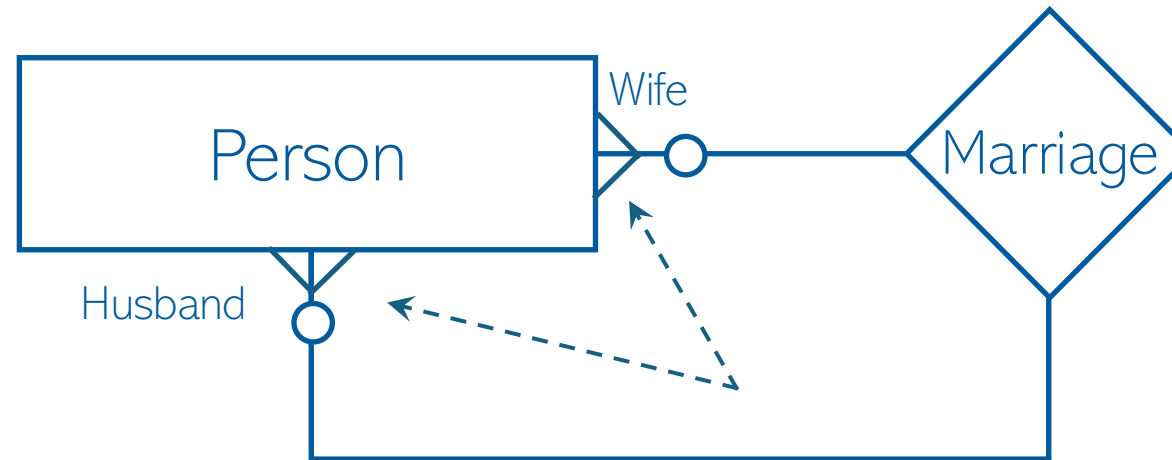
Monogamy 'muh·naa·guh·mee' is the practice having only one partner at a time

Same Gender/Sex Marriage

24

Relationship Set (R): Marriage

A pair of entities of the same entity set have relationship



Sologamy *'suh-log-uh-mee'* or *'sah-luh-guh-mee'* (autogamy) is marriage by a person to themselves

Polygyny *'puh·li·juh·nee'* a marriage where one man has more than one wife **at the same time**

Polyandry *'paa·lee·an·dree'* a form of polygamy where a woman has two or more husbands **at the same time**

Polygamy *'puh·li·guh·mee'* is the practice of being married to more than one spouse **at the same time**

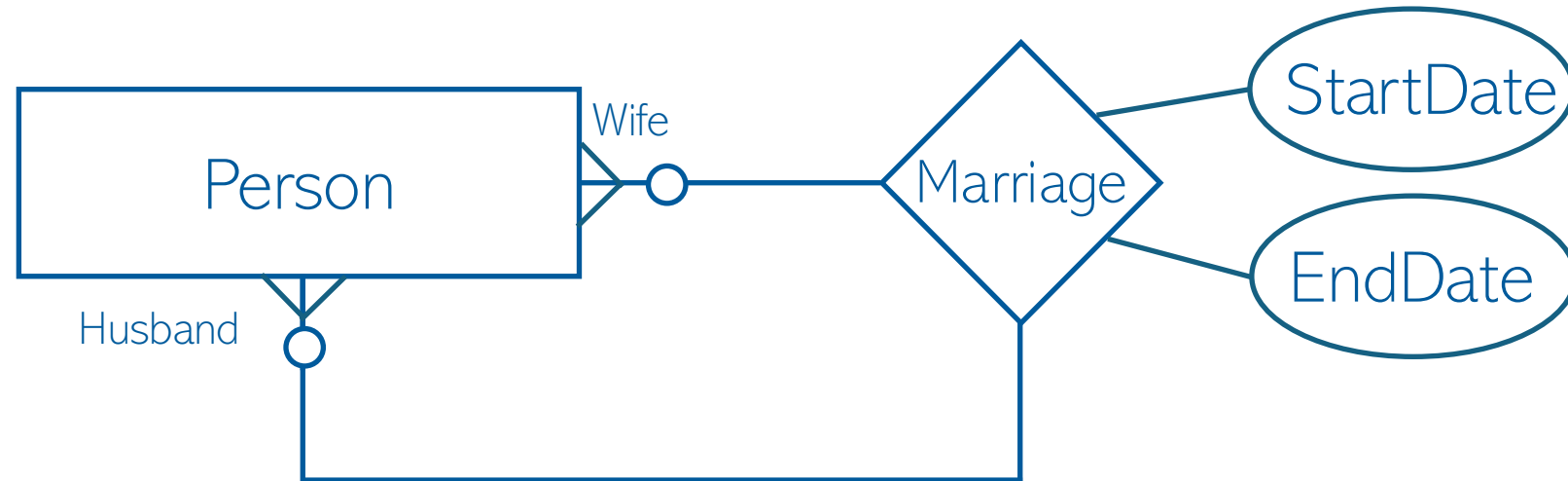
Monogamy *'muh·naa·guh·mee'* is the practice having only one partner at a time

Same Gender/Sex Marriage

25

Relationship Set (R): Marriage

A pair of entities of the same entity set have relationship



Sologamy 'suh-log-uh-mee' or 'sah-luh-guh-mee' (autogamy) is marriage by a person to themselves

Polygyny 'puh·li·juh·nee' a marriage where one man has more than one wife **at the same time**

Polyandry 'paa·lee·an·dree' a form of polygamy where a woman has two or more husbands **at the same time**

Polygamy 'puh·li·guh·mee' is the practice of being married to more than one spouse **at the same time**

Monogamy 'muh·naa·guh·mee' is the practice having **only one partner at a time, yet different times many partner**

ER model cannot explain more or enforce it!

Same Gender/Sex Marriage

27

Identifying Relationship Set for A Weak Entity Sets

By ER principles, an entity set must have a key.

At worst, all attributes are in the key set.

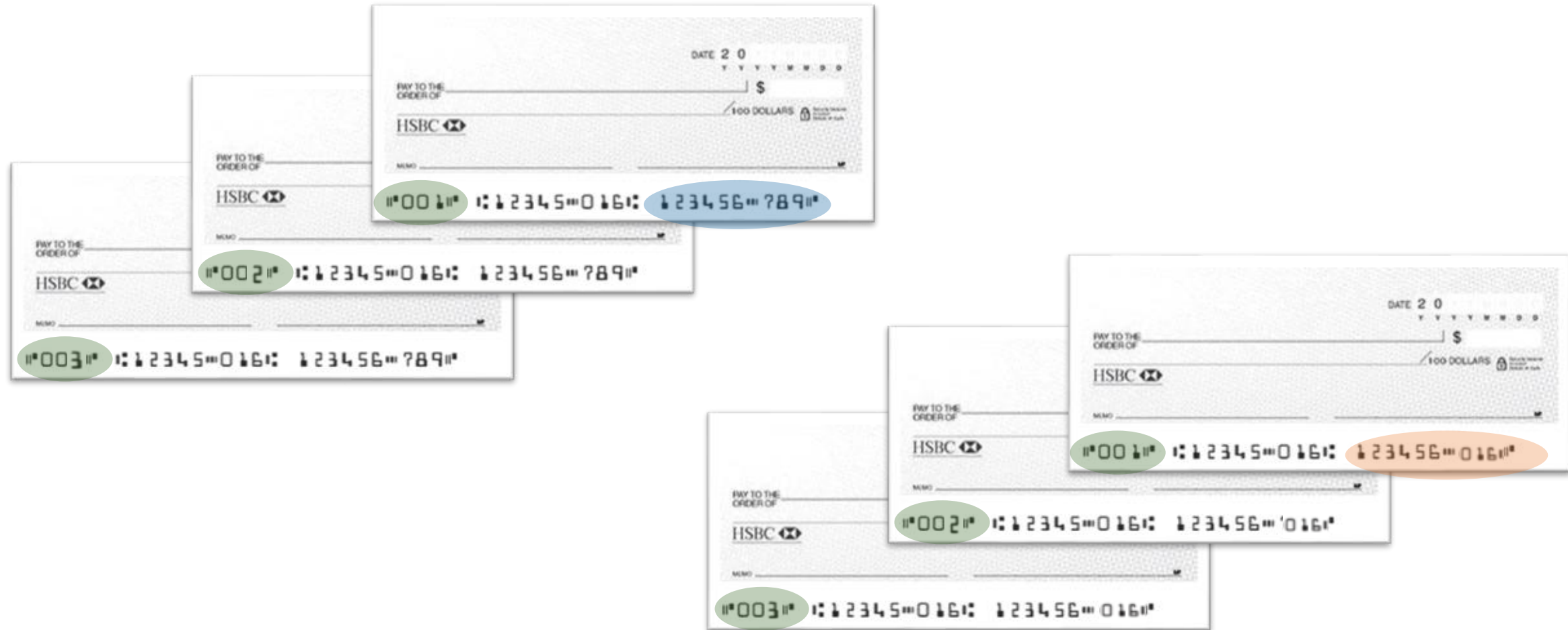
This guarantees **no duplicates** in the entity set.

But, however, tho, ...

28

Identifying Relationship Set for A Weak Entity Sets

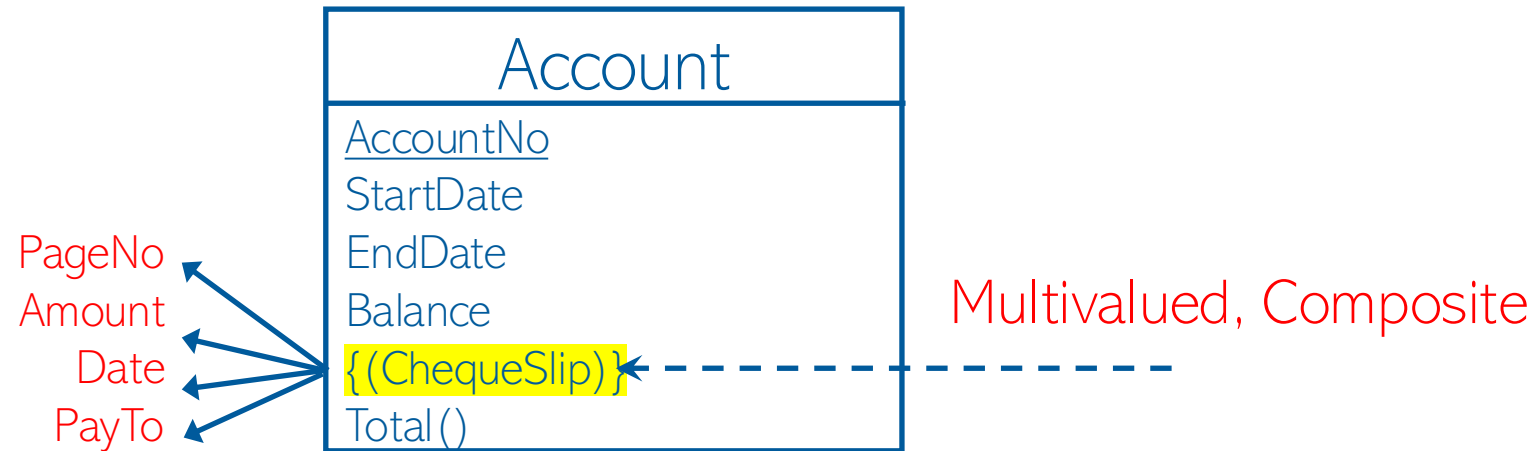
Example from Banking System: Cheque Slips for an Account



29

Identifying Relationship Set for A Weak Entity Sets

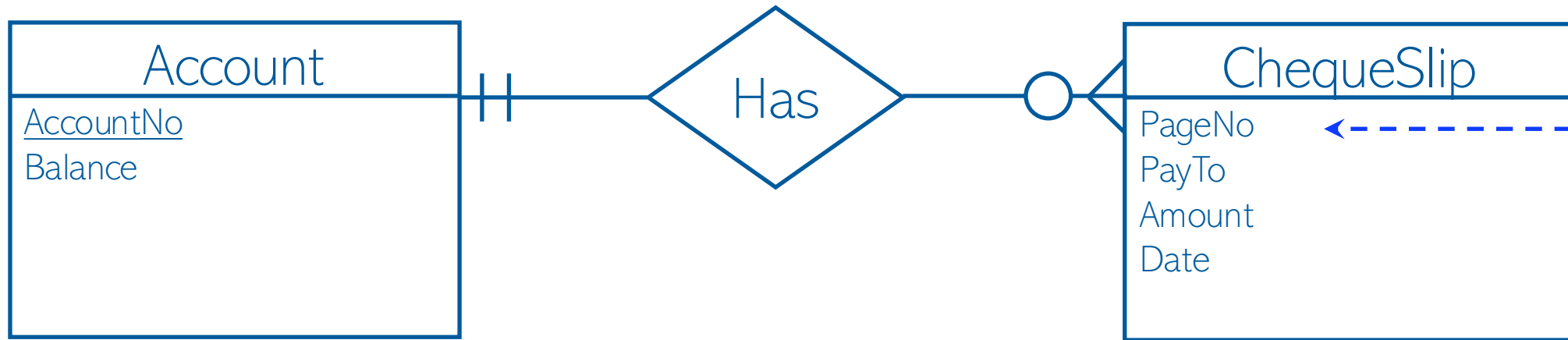
Example from Banking System: Cheque Slips for an Account



30

Identifying Relationship Set for A Weak Entity Sets

Example from Banking System: Cheque Slips for an Account



What is the Primary Key?

Same PageNo, Amount, Date, ..., can be written but from different accounts.

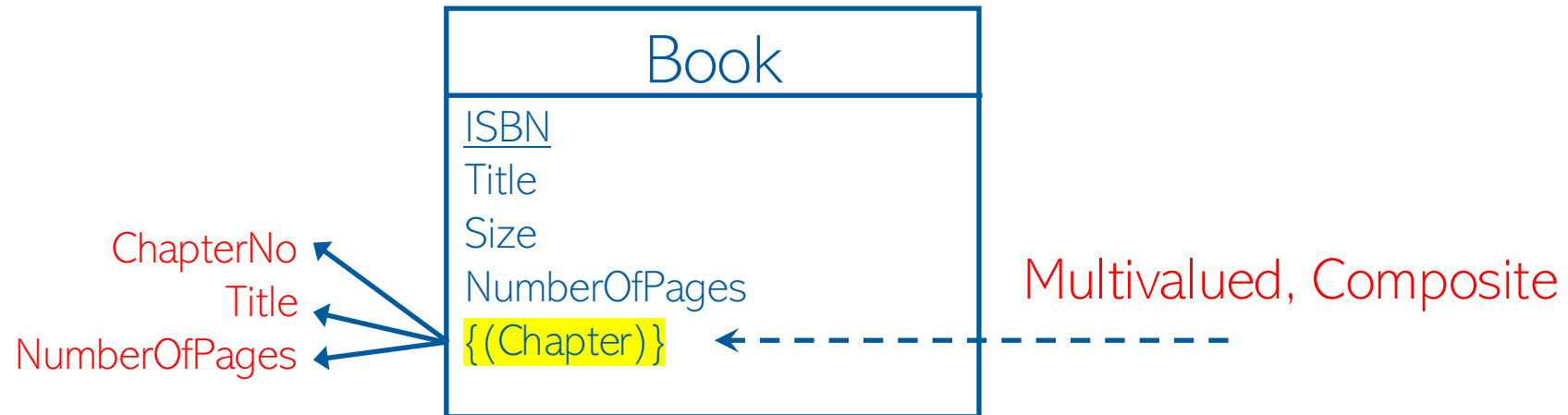
We can have duplicate cheque slips ...



32

Identifying Relationship Set for A Weak Entity Sets

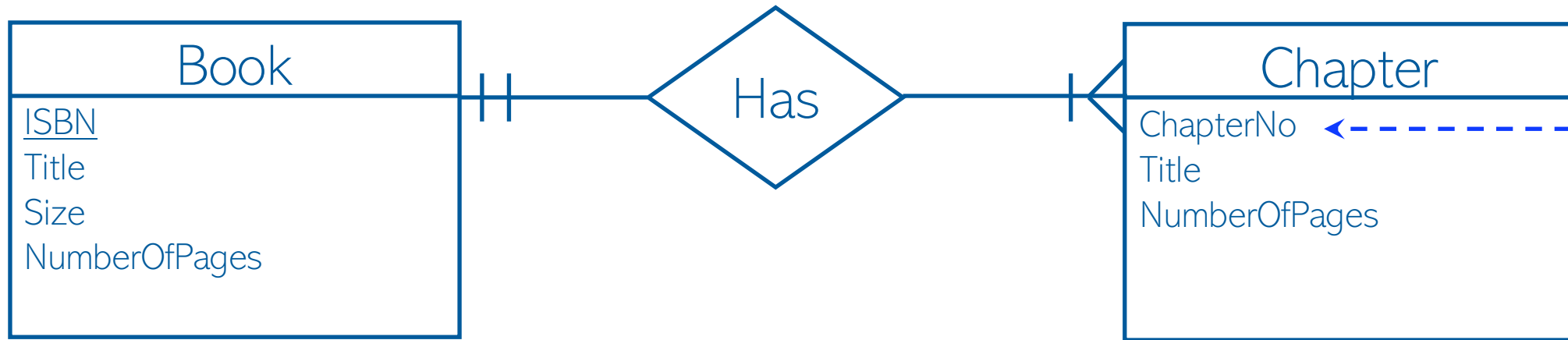
Example from Publication System: Book Chapters for Books



33

Identifying Relationship Set for A Weak Entity Sets

Example from Publication System: Book Chapters for Books



What is the Primary Key?

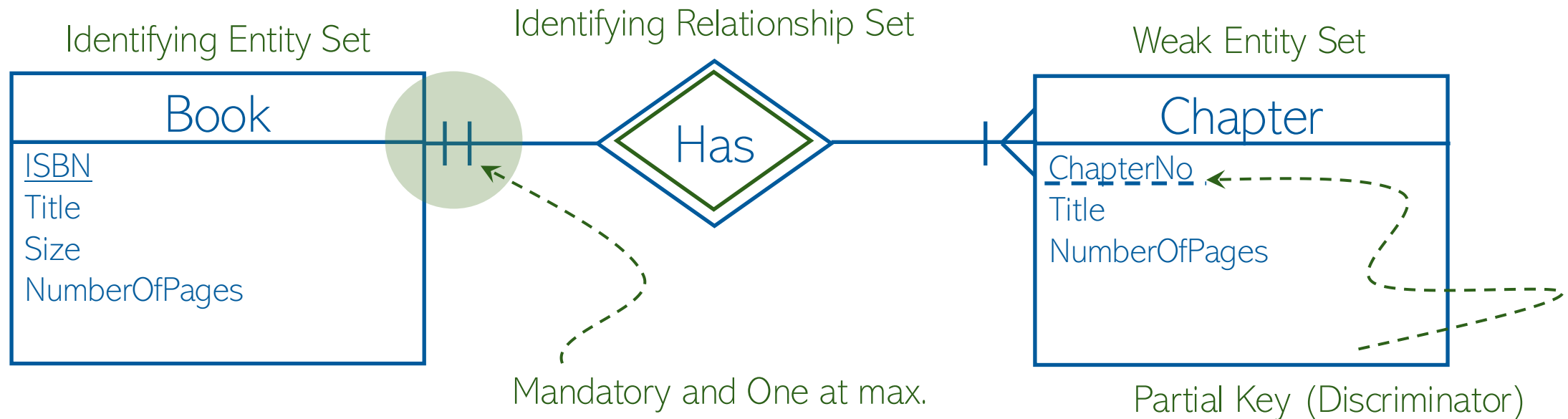
Same ChapterNo, Title, NumberOfPage, ..., can be written but for different books.

We can have duplicate chapters ...

34

Identifying Relationship Set for A Weak Entity Sets

Example from Publication System: Book Chapters for Books



PC-Canada.com

CANADA'S TECHNOLOGY SOURCE SINCE 2003

55 King Street West, Suite 700
Kitchener ON N2G 4W1
Phone: (800) 556-1971
Fax: (888) 849-1760

Invoice

*** 20% RESTOCKING FEE ON RETURNED ITEMS***

ALL WARRANTIES ARE MANUFACTURER'S STATED TERMS AND CONDITIONS

NO RETURNS ACCEPTED WITHOUT PRE-AUTHORIZED APPROVAL

Date

5/7/18

Page

1

Invoice Number

IN620995

Sold To:

HOSSEIN FANI

Ship To:

HOSSEIN FANI

Order No.

1805043501

Order Date

May 4, 2018

Customer No.

Salesperson

PCCAN

Qty. Ord.	Qty. Shp.	Qty. B/O	Item Number	Description
1	1	0	CYB-CRA50005	CYBERPOWER CARBON CRA50005 RACK SHELF - 19 1U WIDE X 32.70
			PCFRT	SHIPPING FEE

Comments:

HST 89527 0213 RT001
QST 1202169020

Tax summary:

For our return policy, please see www.PC-C

PC-Canada.com

CANADA'S TECHNOLOGY SOURCE SINCE 2003

55 King Street West, Suite 700
Kitchener ON N2G 4W1
Phone: (800) 556-1971
Fax: (888) 849-1760

Invoice

*** 20% RESTOCKING FEE ON RETURNED ITEMS***

ALL WARRANTIES ARE MANUFACTURER'S STATED TERMS AND CONDITIONS

NO RETURNS ACCEPTED WITHOUT PRE-AUTHORIZED APPROVAL

Due Date

PAID BY CREDIT CARD

Date

5/9/18

Page

1

Invoice Number

IN621412

Sold To:

HOSSEIN FANI

Ship To:

HOSSEIN FANI

Order No.	Order Date	Customer No.	Salesperson	PO Number	Ship Via	Terms
1805091667	May 9, 2018		PCCANADA			Credit Card

Qty. Ord.	Qty. Shp.	Qty. B/O	Item Number	Description	Unit Price	UOM	Extended Price
1	1	0	CYB-CRA50005	CYBERPOWER CARBON CRA50005 RACK SHELF - 19 1U WIDE X 32.70	90.99	EACH	90.99
			PCFRT	SHIPPING FEE			48.99

Comments:

HST 89527 0213 RT001
QST 1202169020

Tax summary:

Subtotal

Total sales tax

Total amount

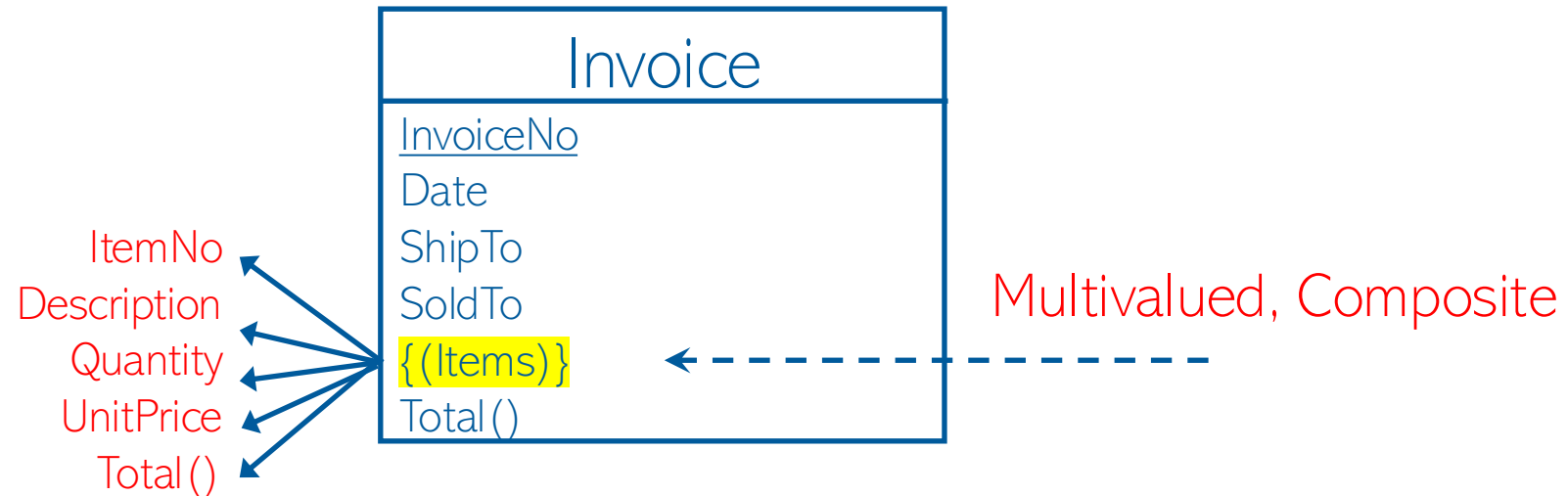
Paid in Full

For our return policy, please see www.PC-Canada.com

36

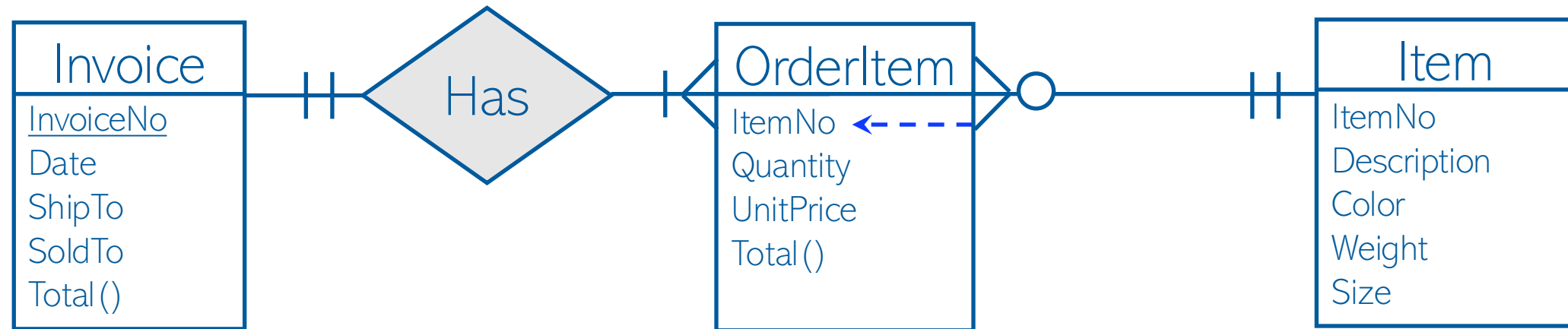
Identifying Relationship Set for A Weak Entity Sets

Example from Publication System: Book Chapters for Books



37

Identifying Relationship Set for A Weak Entity Sets



What is the Primary Key?

Same item, same quantity, ..., can be sold in different purchases.

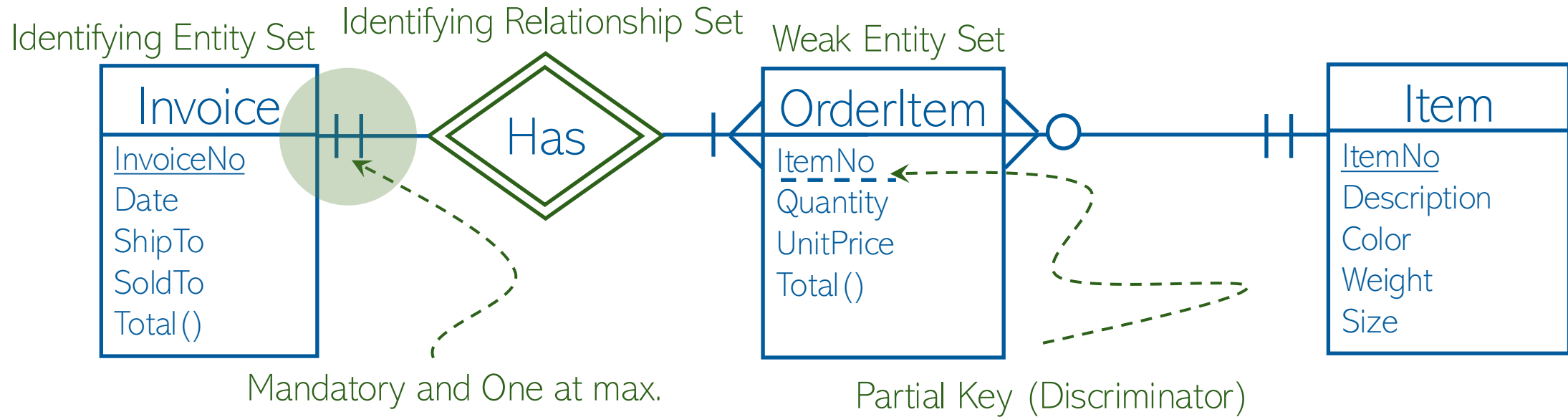
We can have duplicate items ...

Ord.	Shp.	B/O	Item Number	Description	Unit Price	UOM	Extended Price
1	1	0	CYB-CRA50005	CYBERPOWER CARBON CRA50005 RACK SHELF - 19 1U WIDE X 32.70	90.99	EACH	90.99
			PCFRT	SHIPPING FEE			

Ord.	Shp.	B/O	Item Number	Description	Unit Price	UOM	Extended Price
1	1	0	CYB-CRA50005	CYBERPOWER CARBON CRA50005 RACK SHELF - 19 1U WIDE X 32.70	90.99	EACH	90.99
			PCFRT	SHIPPING FEE			48.99

38

Identifying Relationship Set for A Weak Entity Sets



Date	Page
5/7/18	1
Invoice Number IN620995	



Ord.	Shp.	B/O	Item Number	Description	Unit Price	UOM	Extended Price
1	1	0	CYB-CRA50005	CYBERPOWER CARBON CRA50005 RACK SHELF - 19 1U WIDE X 32.70	90.99	EACH	90.99
			PCFRT	SHIPPING FEE			48.99

Date	Page
5/9/18	1
Invoice Number IN621412	



Ord.	Shp.	B/O	Item Number	Description	Unit Price	UOM	Extended Price
1	1	0	CYB-CRA50005	CYBERPOWER CARBON CRA50005 RACK SHELF - 19 1U WIDE X 32.70	90.99	EACH	90.99
			PCFRT	SHIPPING FEE			48.99

Identifying Relationship Set for A Weak Entity Sets

By ER principles, an entity set must have a key.

At worst, all attributes are in the key set.

This guarantees no duplicates in the entity set.

But, however, tho, ...

Weak entity set does *not* have key but “*partial*” key. The key set is formally:

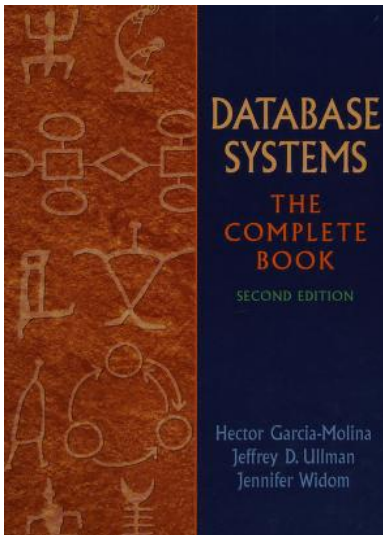
$$\{\text{key set of the identifying entity set}\} \cup \{\text{partial key set}\}$$

ER Design Principles

Faithfulness:	e.g., entity set, attribute according to system/domain.
Redundancy:	e.g., circular relationship sets
Simplicity:	e.g., single attribute key



Outline: W03B



CH04 (2nd Ed.)

I) Entity

II) Attribute

III) Relationship

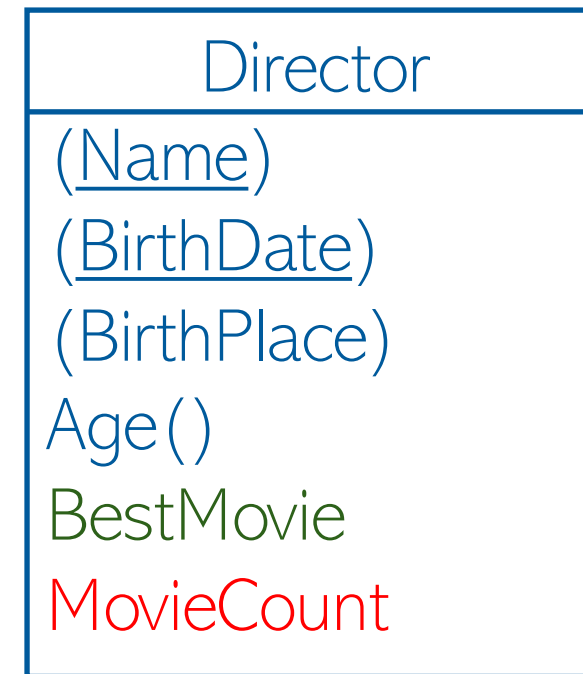
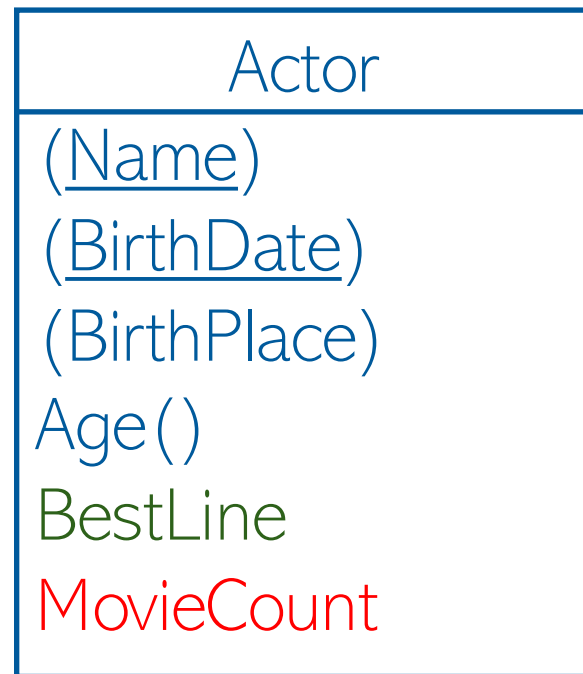
III) Extended ER (EER)



Conceptual Level | Extended Entity-Relationship Model

IS-A (Inheritance, Generalization $\leftrightarrow$ Specification)

Both are person, so they have person's attributes like name, age, ...

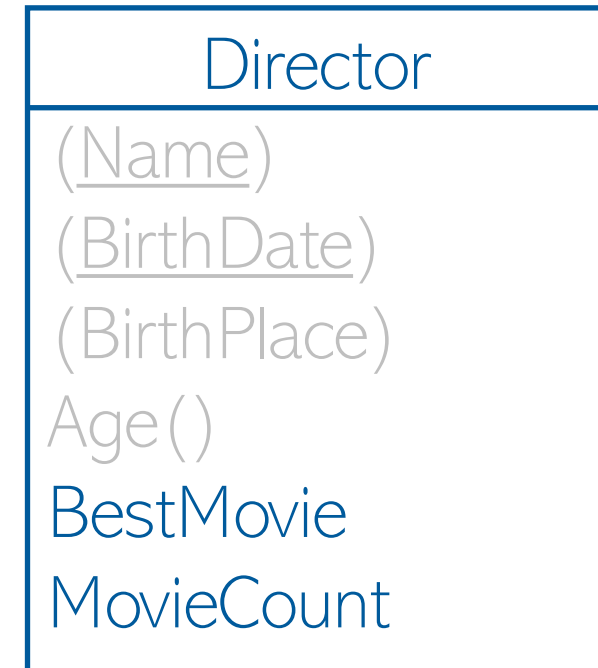
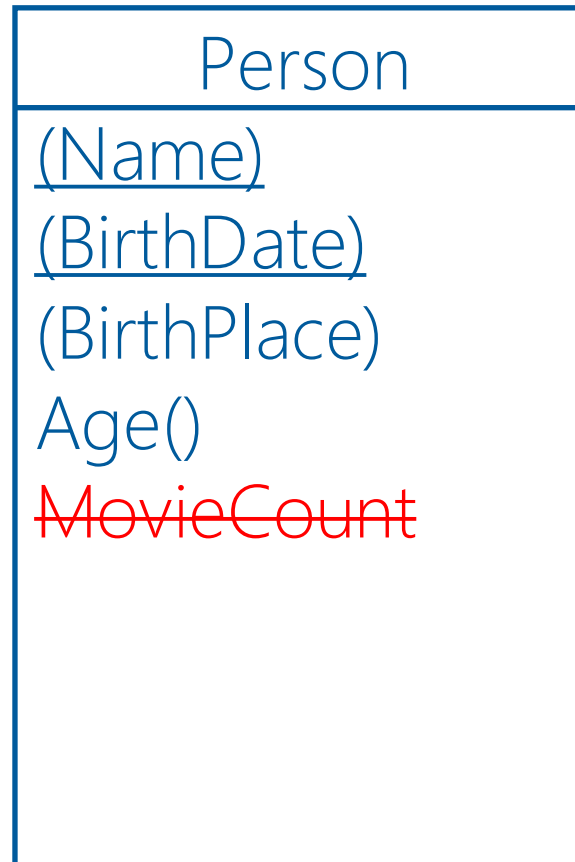
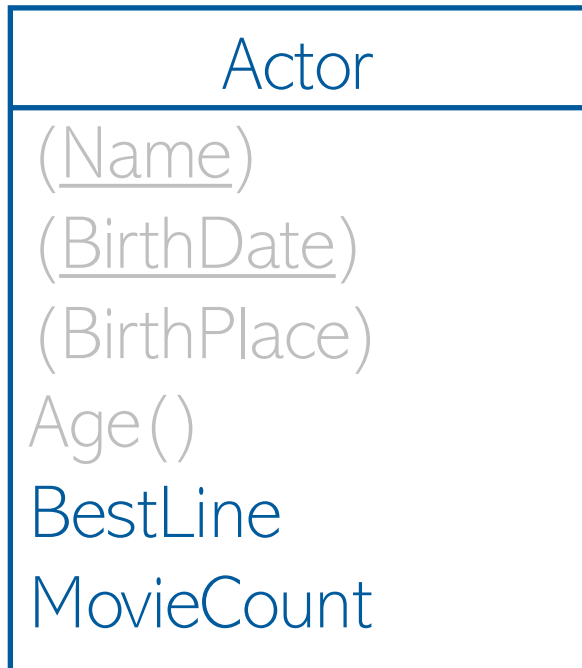


52

IS-A (Inheritance, Generalization $\leftrightarrow$ Specification)

Both are person, so they have person's attributes like Name, Age, ...

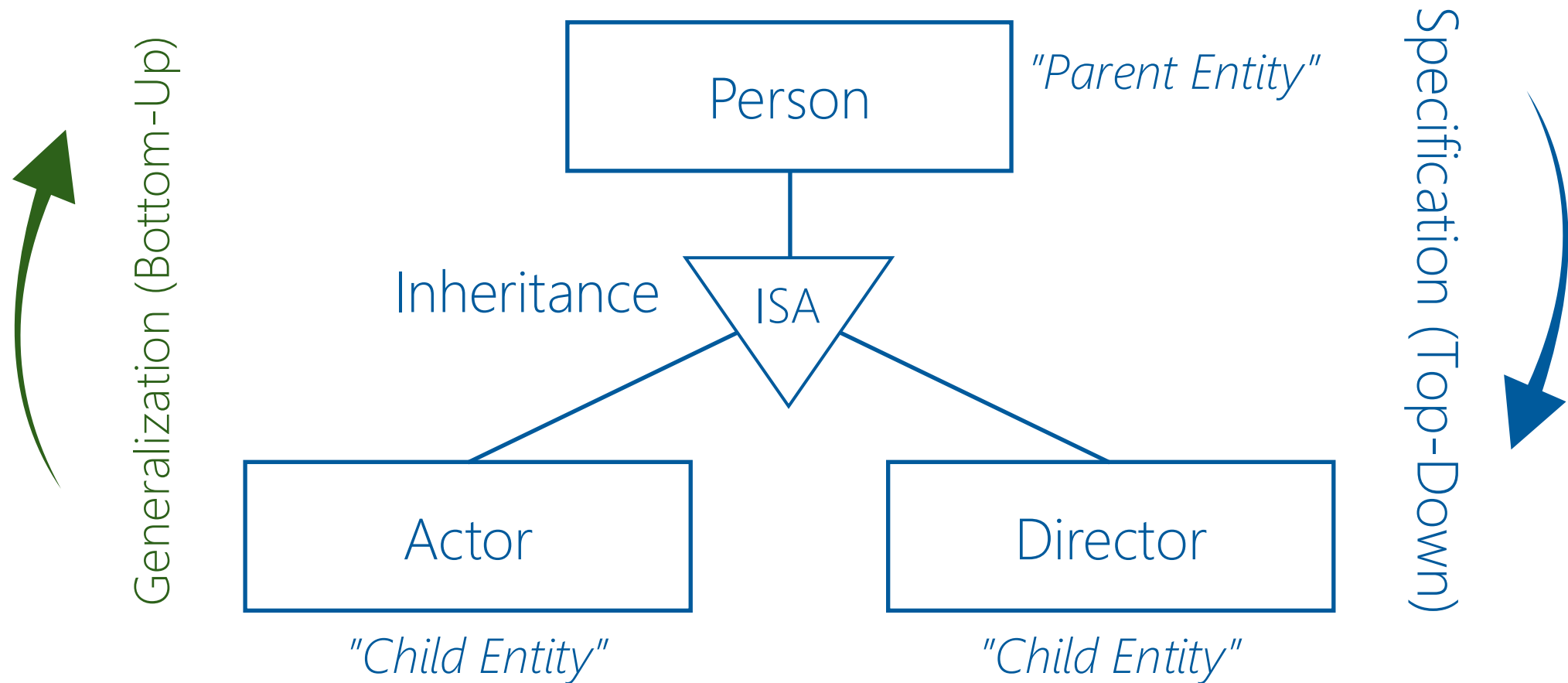
MovieCount, while has same attribute name, it has different semantics!



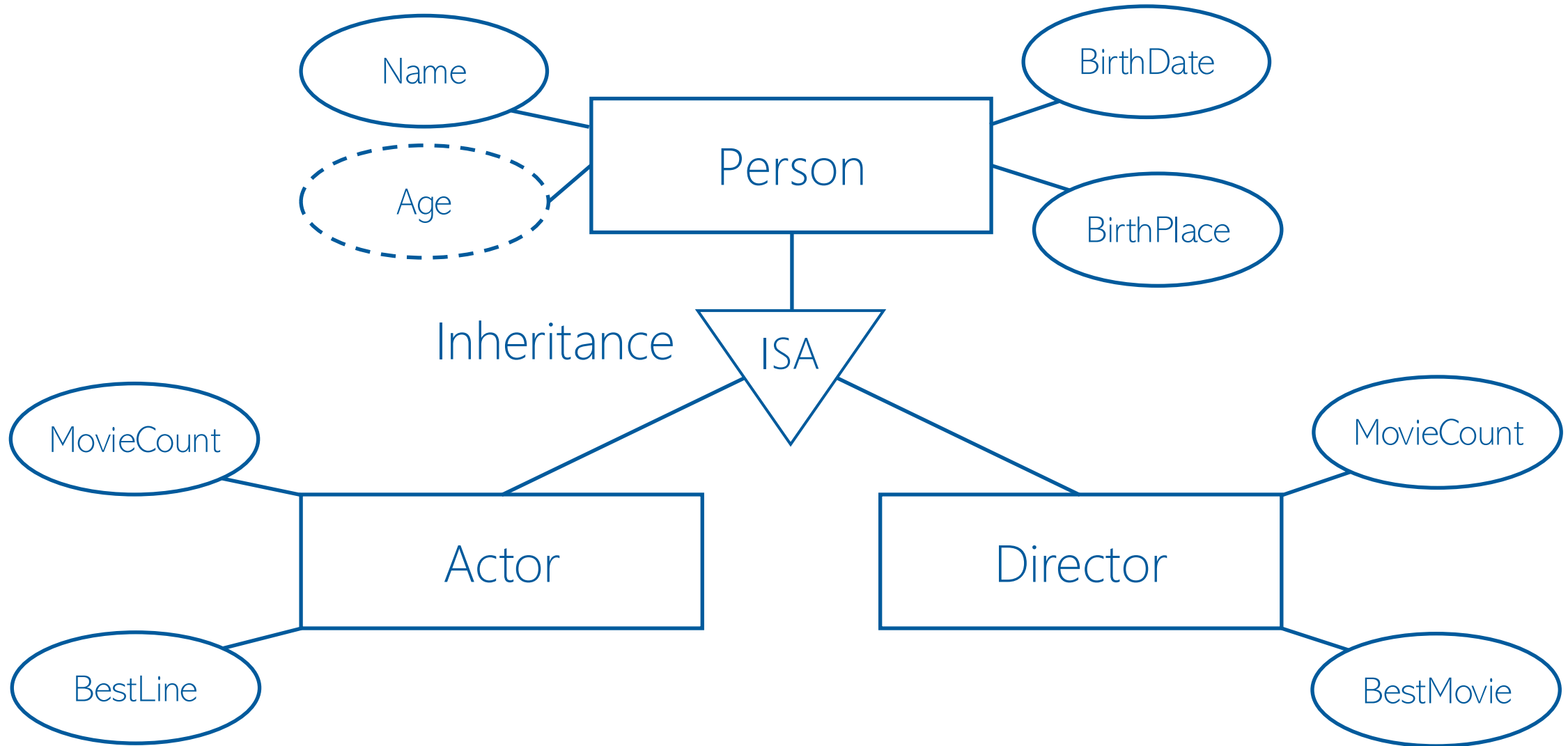
53

IS-A (Inheritance, Generalization $\leftrightarrow$ Specification)

Both are person, so they have person's attributes like name, age, ...



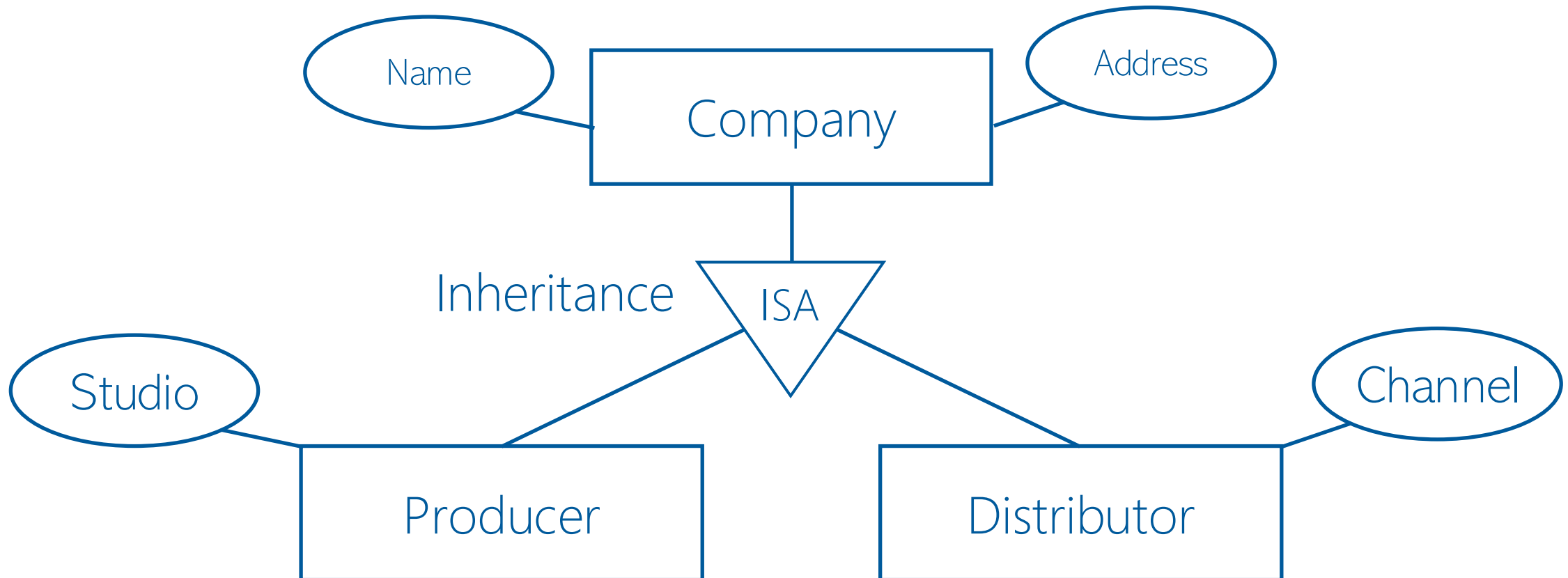
IS-A (Inheritance, Generalization $\leftrightarrow$ Specification)



55

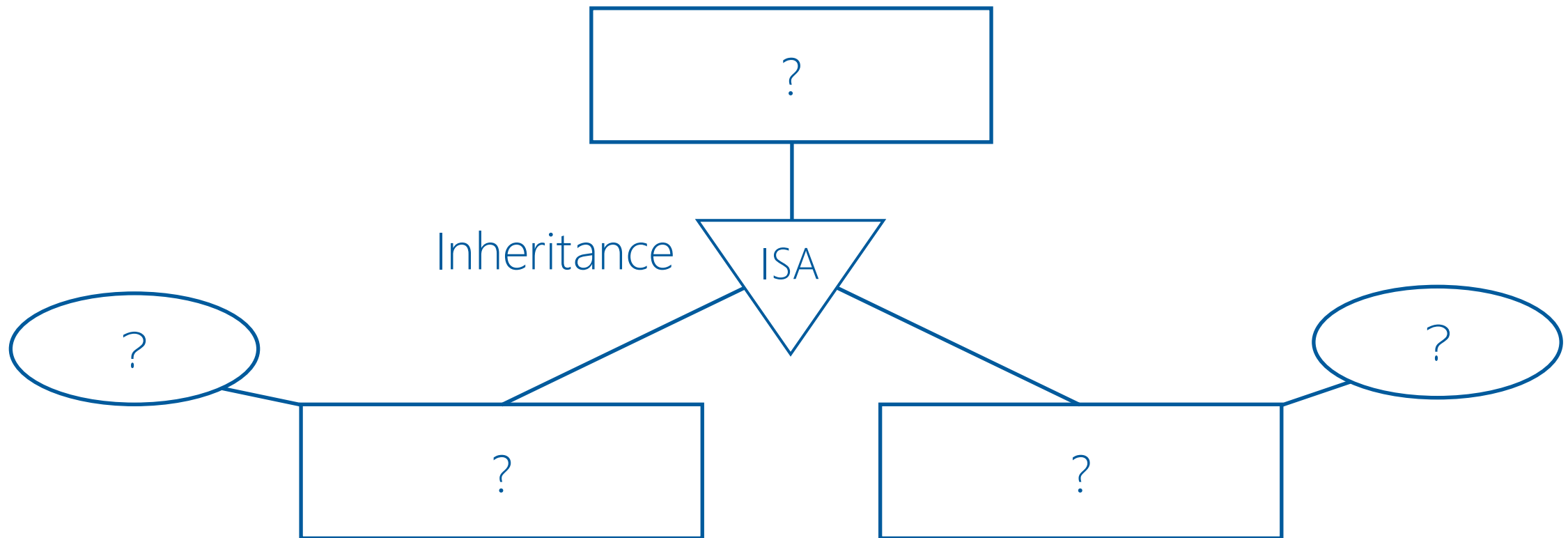
IS-A (Inheritance, Generalization $\leftrightarrow$ Specification)

Both are Company, so they have Company's attributes like name, address, ...



IS-A (Inheritance, Generalization $\leftrightarrow$ Specification)

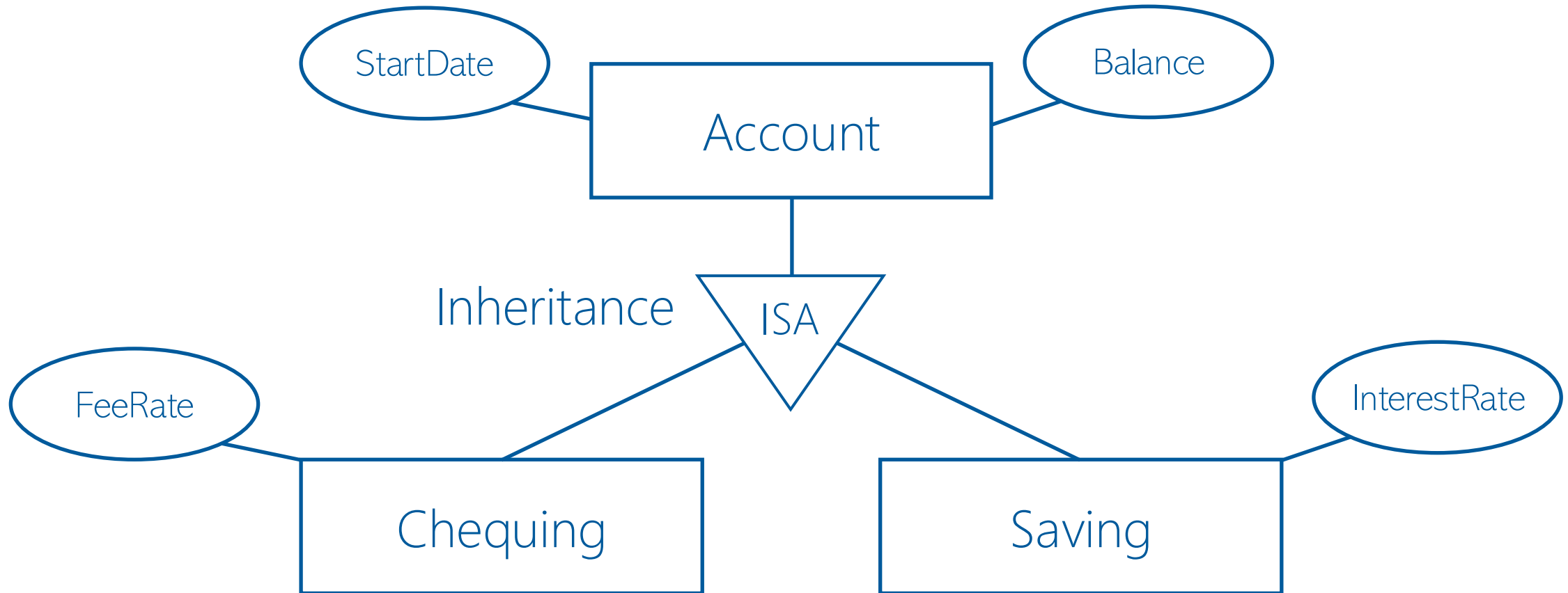
In a banking system, ...



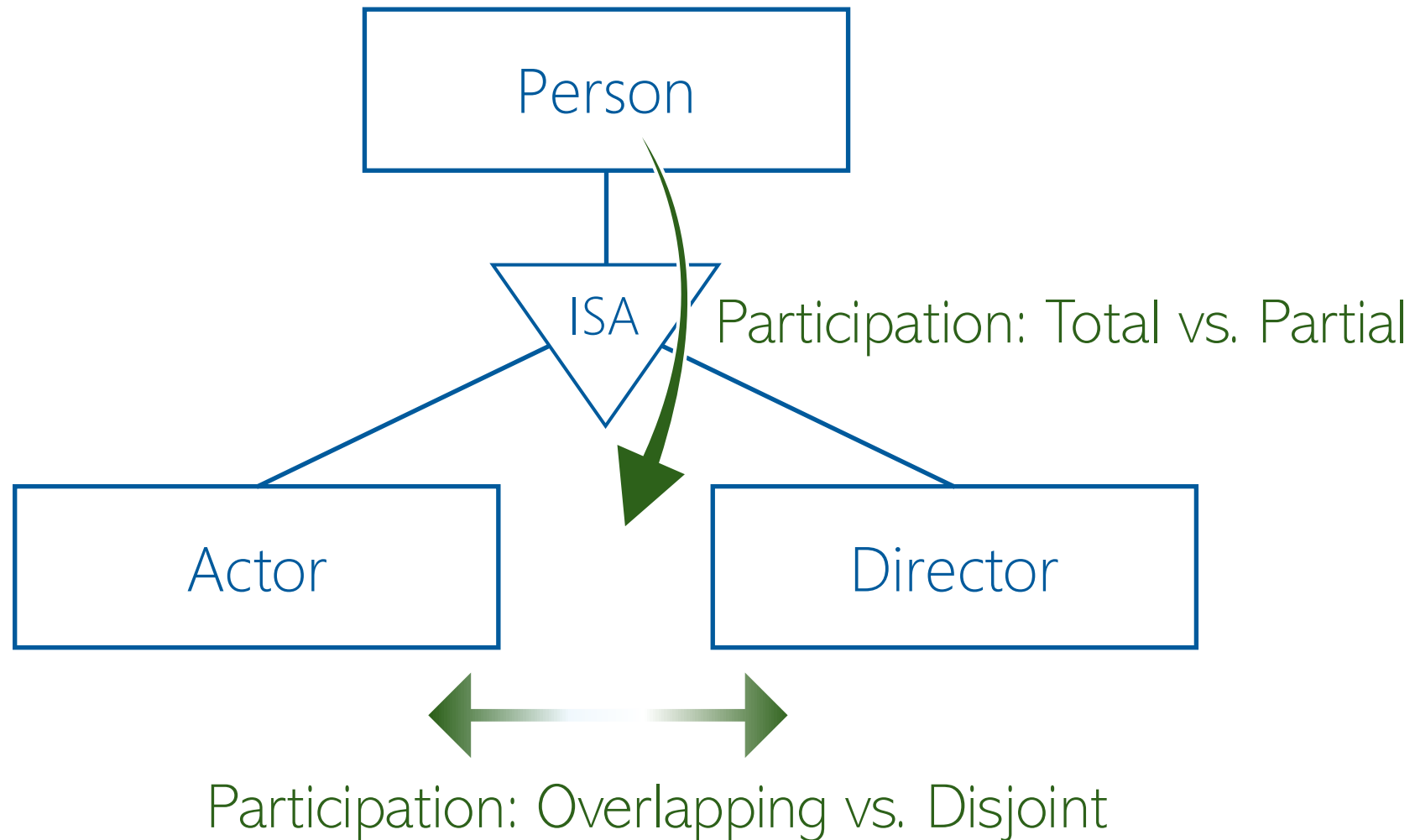
57

IS-A (Inheritance, Generalization $\leftrightarrow$ Specification)

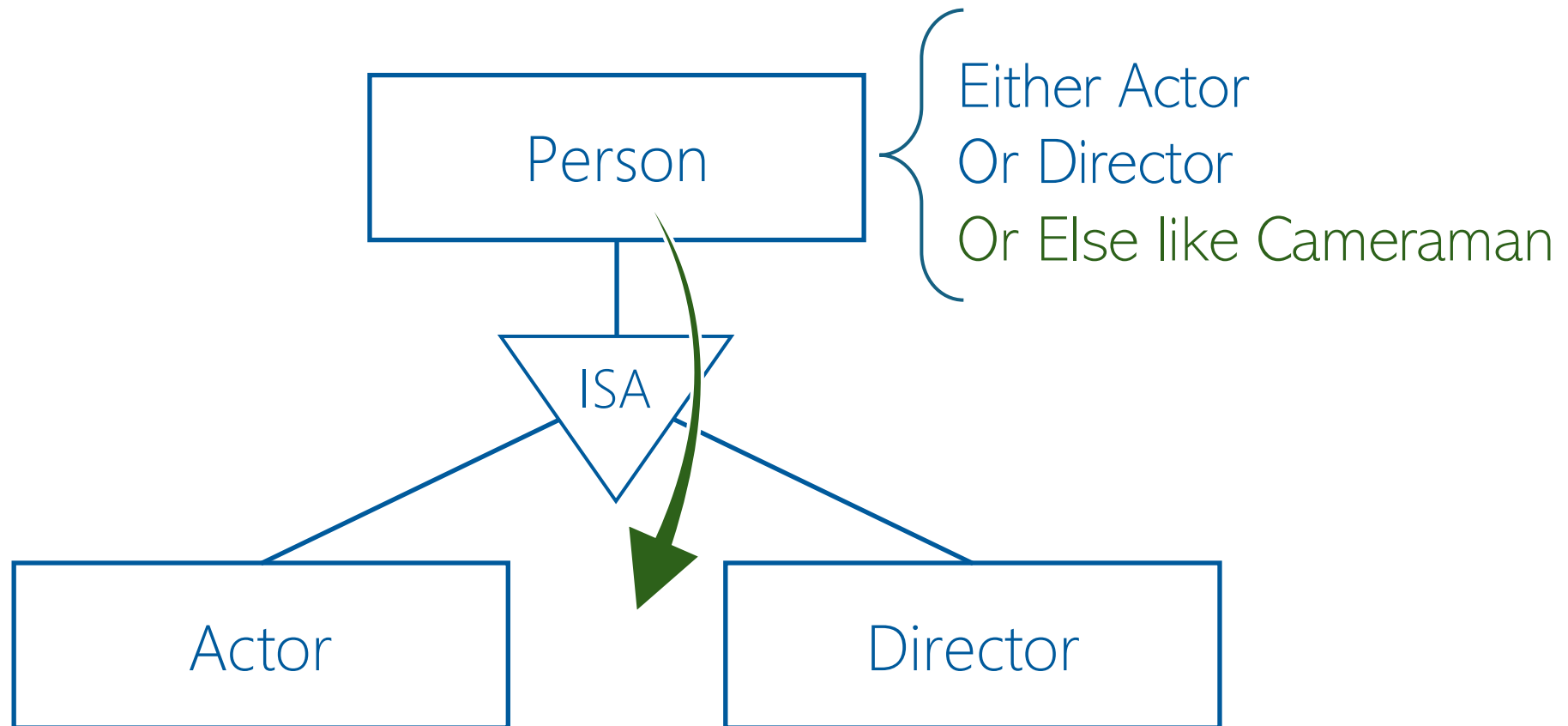
In a banking system, ...



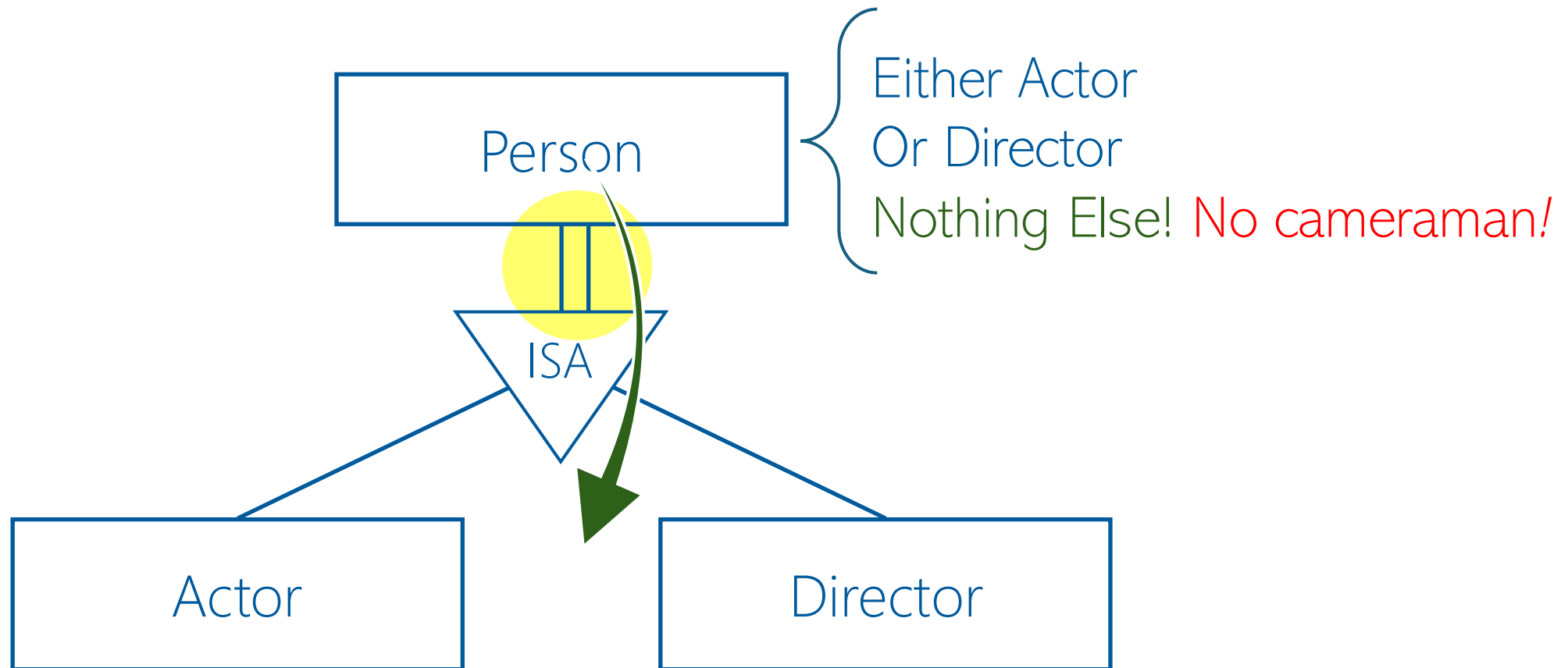
IS-A: No Ordinality Nor Cardinality but Constraints



IS-A: Partial Participation (Default)

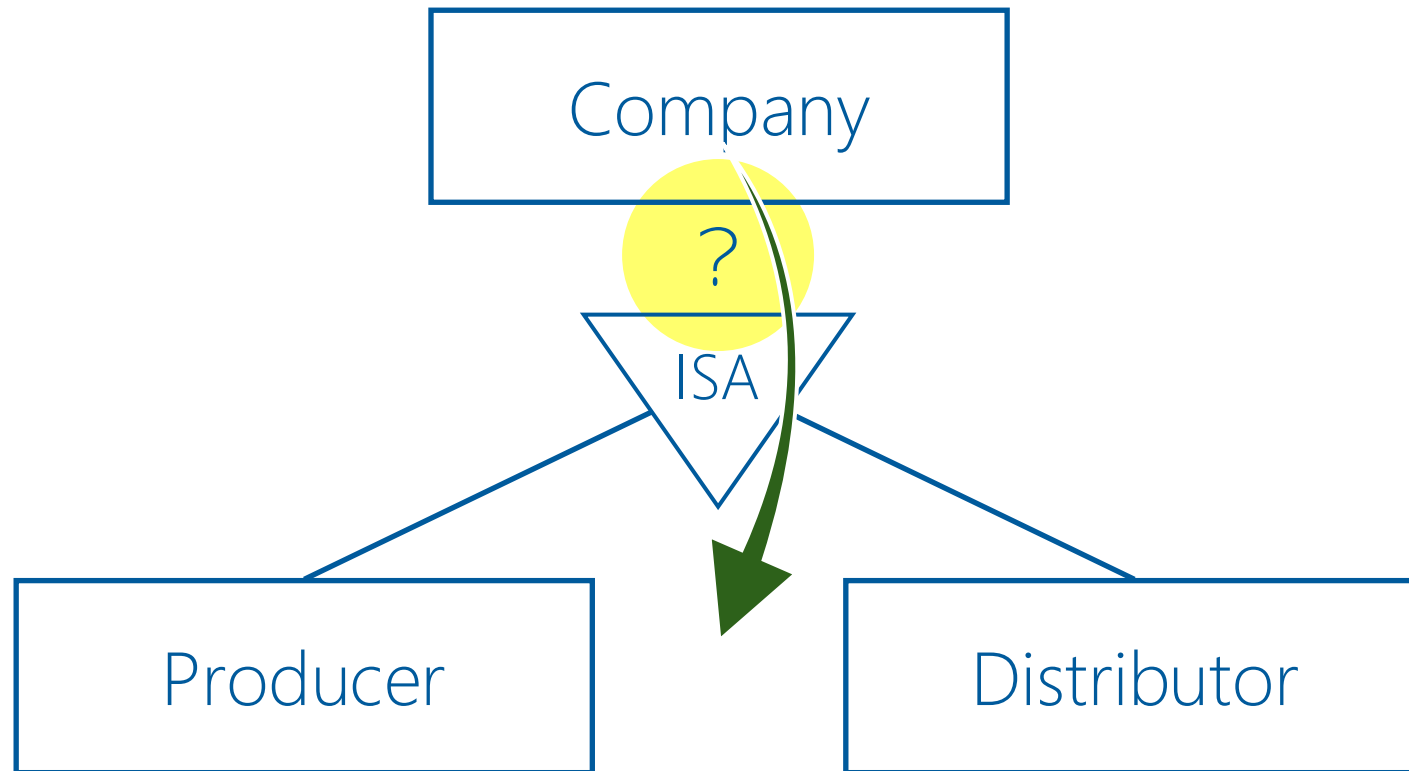


IS-A: Total Participation

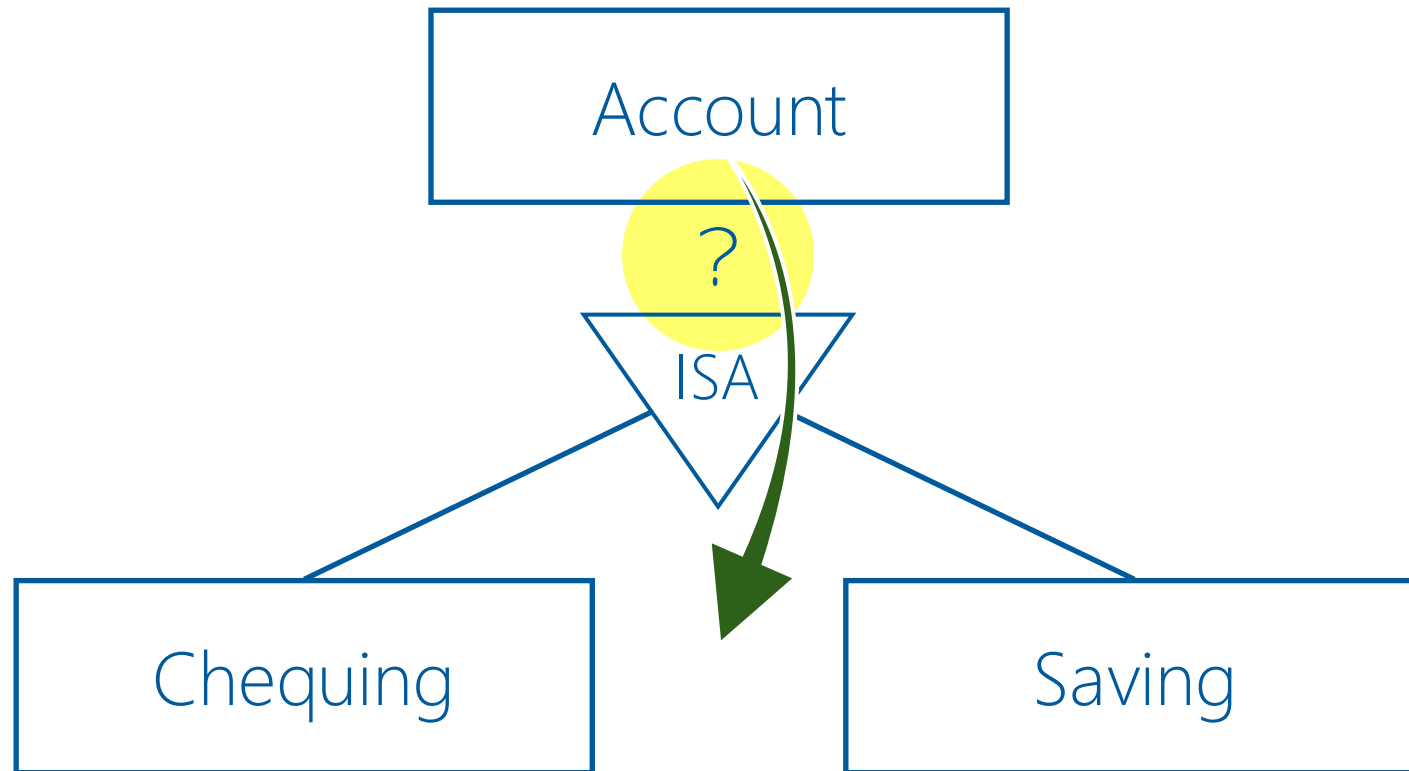


61

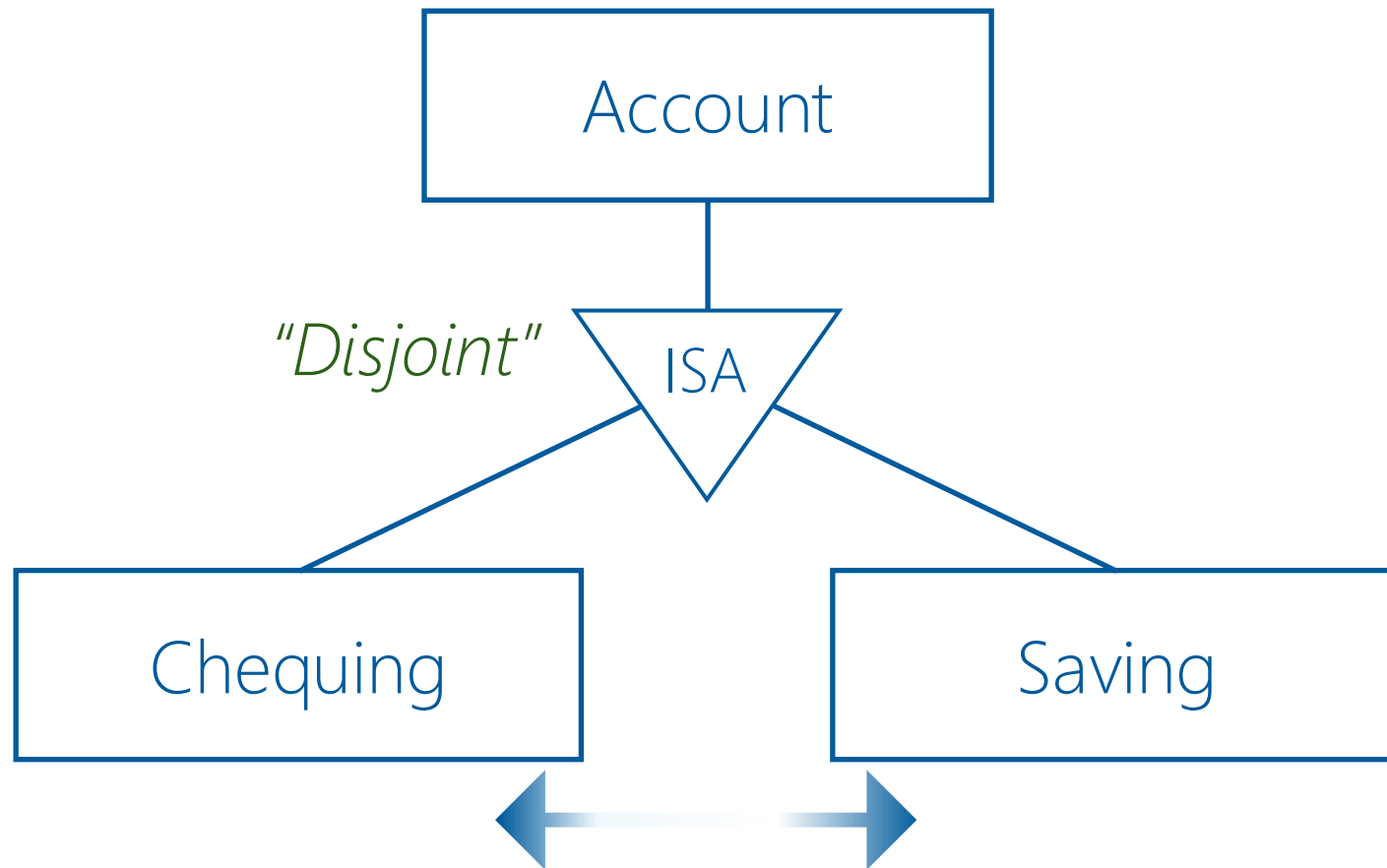
IS-A: Total vs. Partial Participation



IS-A: Total vs. Partial Participation

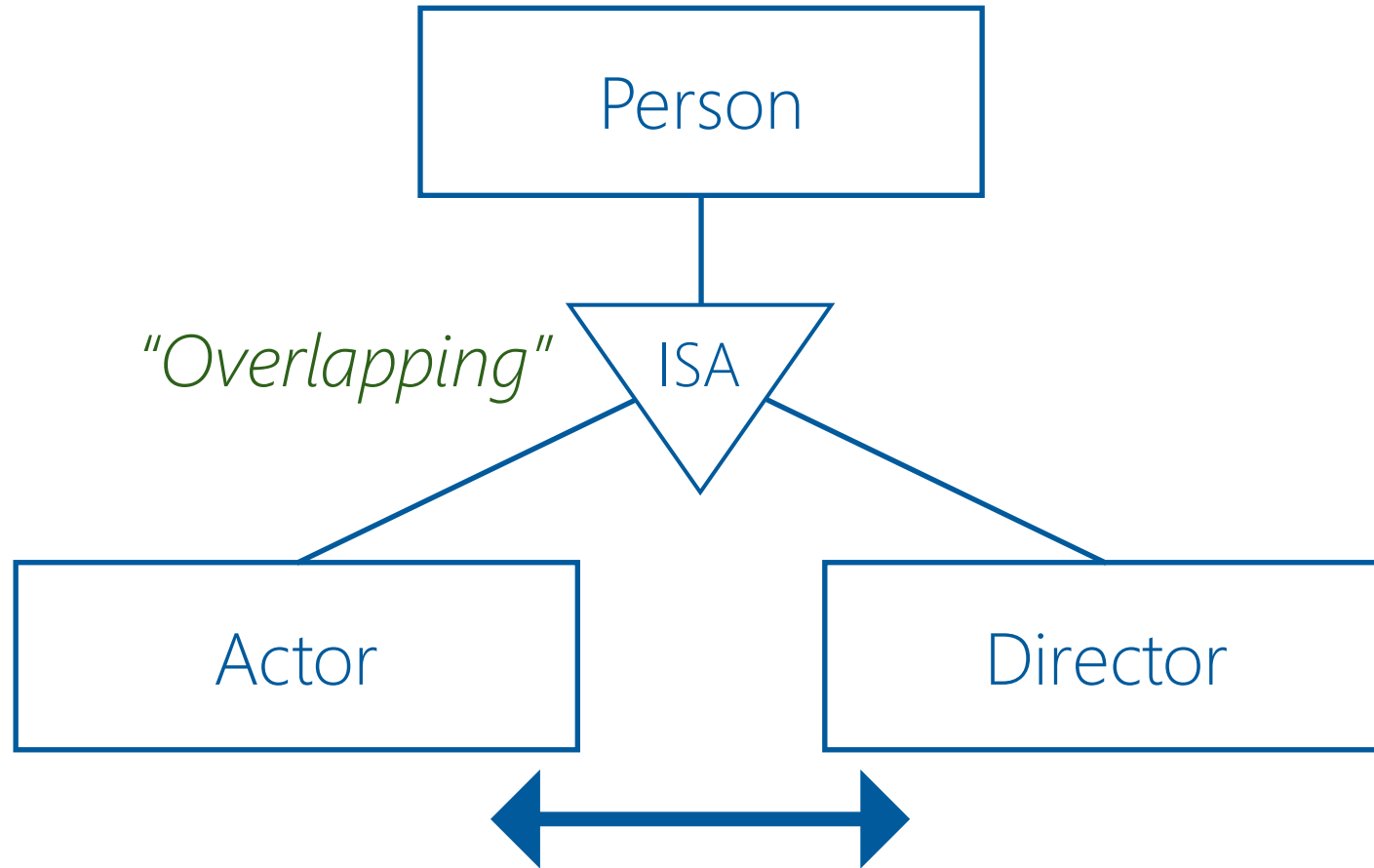


63 IS-A: Disjoint Participation



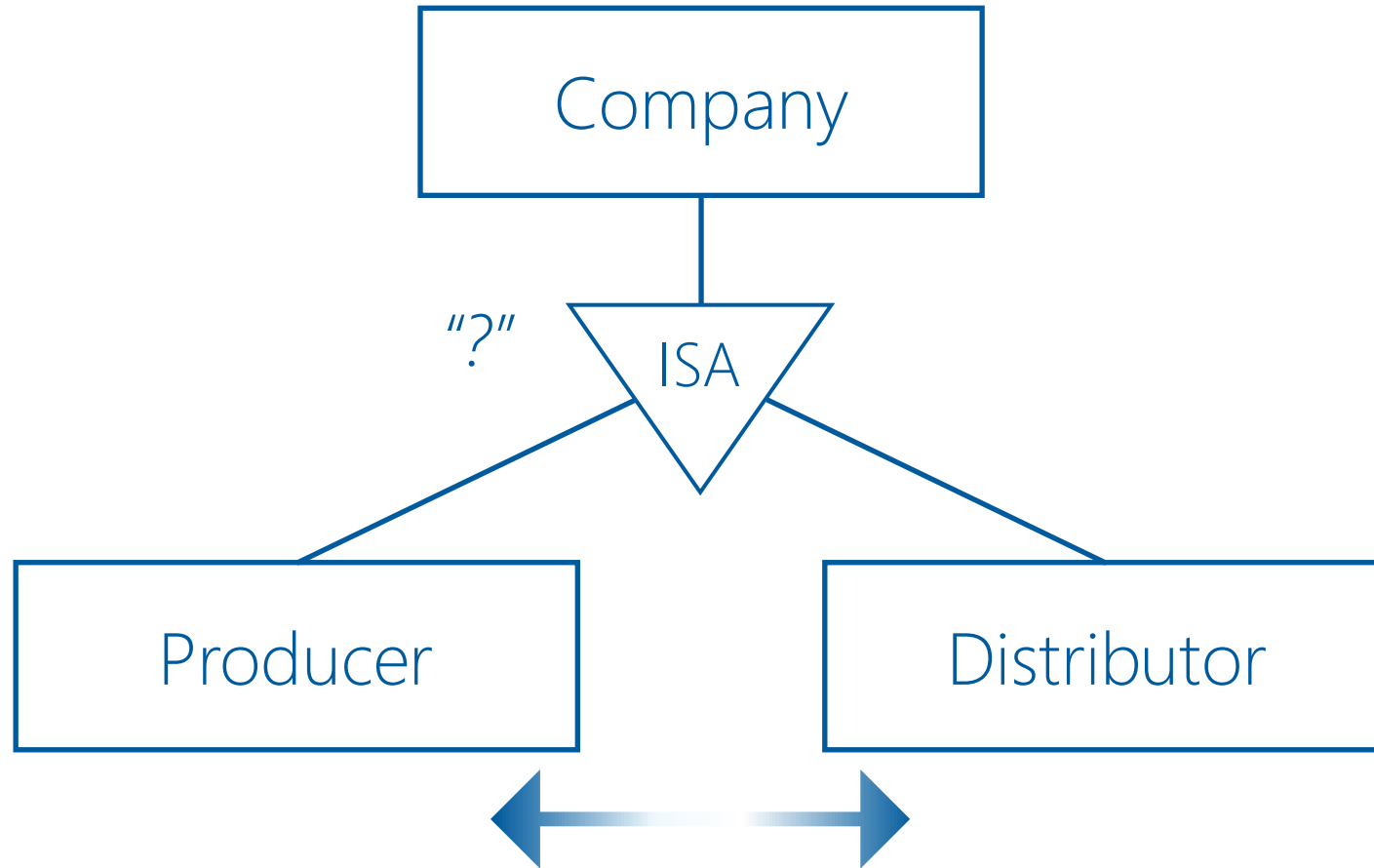
An Account can be either Chequing or Saving **but not both**.

IS-A: Overlapping Participation



A Person can be both Actor and Director.

IS-A: Disjoint vs. Overlapping Participation

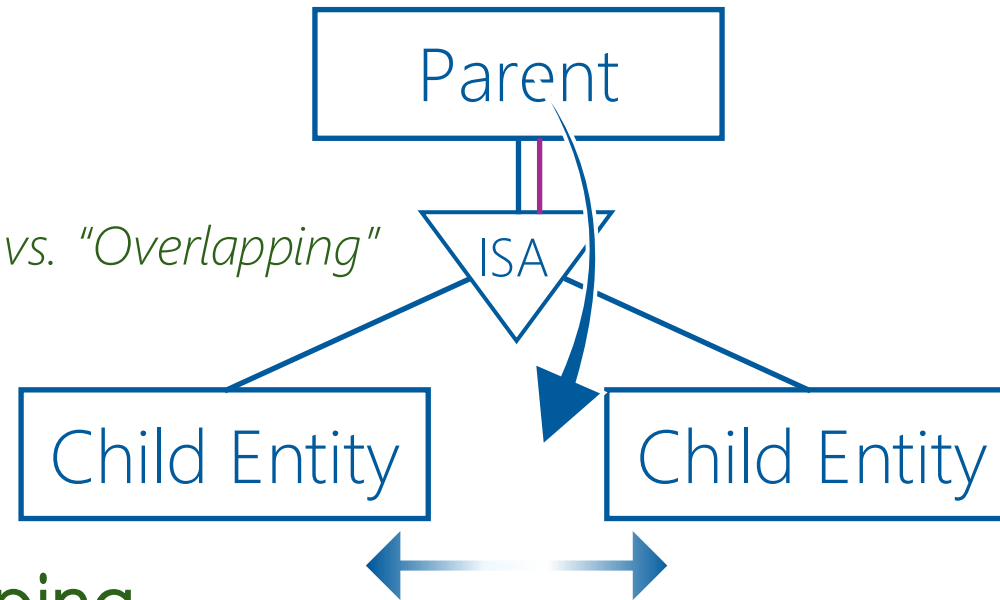


A Company can be both Producer and Distributor? Or not?

IS-A: Participation

Parent: Partial vs. Total

"Disjoint" vs. "Overlapping"



Children: Disjoint vs. Overlapping



MR. HITCHCOCK

68

A Data Odyssey: Relational DBMS vs. GGG



Data Rep.
in
RDBMS

Real World Entity

Conceptual Level | Entity-Relationship Model Level

| Logical Level | Relational Model

| Physical Level | SQL

| Computable Entity



Data Rep.
in
GGG

Real World Entity

Conceptual Level | Web Ontology Language (OWL)

| Logical Level | RDF & Description Language

| Physical Level | SPARQL

| Computable Entity

69

A Data Odyssey: Relational DBMS vs. GGG



Data Rep.
in
RDBMS

Real World Entity

Conceptual Level | Entity-Relationship Model Level

| Logical Level | Relational Model

| Physical Level | SQL

| Computable Entity



Data Rep.
in
GGG

Real World Entity

Conceptual Level | Web Ontology Language (OWL)

| Logical Level | RDF & Description Language

| Physical Level | SPARQL

| Computable Entity